PBiGAMP-Based Joint Channel Estimation and Data Detection Under Nonuniform Doppler Shifts

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***Abstract*—Achieving reliable underwater acoustic communication in time-varying channels characterized by nonuniform Doppler shifts across different channel paths due to variations in arrival and departure angles is a challenging task. In such a scenario, a short package is preferred as the channel parameters, e.g., amplitudes, delays and Doppler shifts can be assumed quasi-static within its transmission duration. By modeling the residual Doppler frequency of each channel tap as a continuous parameter, an input-output relationship for the resampled received signal is formulated. Building on the parametric bilinear generalized approximate message passing (PBiGAMP) framework, we develop an iterative receiver that employs Newton's method for gridless Doppler frequency estimation, thereby enabling joint estimation of channel amplitudes, Doppler frequencies, and data symbol decoding. We refer to this approach as PBiGAMP for nonuniform Doppler shifts (PBiGAMP-NDS). The performance of PBiGAMP-NDS is validated via numerical simulations and sea trials. Notably, for simulation channels with nonuniform Doppler shifts—where the standard deviation of relative moving speed related to channel path σ_v is 0.05 m/s and the minimum time delay between paths with significant Doppler differences normalized by the symbol duration N_τ is 2, when coded bit error rate (BER) is 10^{-3}, PBiGAMP-NDS provides a 2.5 dB signal-to-noise ratio (SNR) gain over the original PBiGAMP. In sea trials, its coded BER is found to be an order of magnitude lower than that of the original PBiGAMP.**

***Index Terms*—Gridless Doppler frequency estimation, joint channel estimation (CE) and equalization, message passing, nonuniform Doppler shifts, single carrier underwater acoustic (UWA) communication.**

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I. Introduction

UNDERWATER acoustic (UWA) communication systems employing phase coherent demodulation arise in a wide range of applications. However, their efficacy is substantially compromised by the inherent complexities of UWA channels. The primary impediments inherent in UWA channels include the following.

1) Extended delay spreads in UWA channels introduce severe inter-symbol interference (ISI).
2) The low propagation speed, transceiver platform mobility, and medium fluctuations jointly induce dilation/compression of the received signal.
3) Individual path-dependent Doppler shifts arise from variations in departure and arrival angles [1].

To mitigate ISI, two primary equalization strategies are typically employed: direct adaptation (DA) and channel estimation (CE)-based equalizations. DA-based methods, exemplified by the recursive least squares (RLS) decision feedback (DFE) algorithms [2] and the improved proportionate normalized minimum-symbol error rate algorithm [3], necessitate extensive pilot sequences to attain filter convergence. Conversely, CE-based approaches, encompassing variants of the minimum mean squared error (MMSE) DFE equalizer [4] and iterative block DFE in the frequency domain [5], typically demand shorter pilot sequences but entail higher computational complexity. To further optimize the system performance while reducing the computation complexity, a joint CE and decoding receiver [6] based on the parametric bilinear generalized approximate message passing (PBiGAMP) algorithm [7] is proposed, demonstrating substantial performance gains under linear time invariant (LTI) channels.

To mitigate Doppler-induced signal dilation/compression, numerous approaches have been proposed, including phase locked loop [2], cross-correlation-based resampling [8], adaptive loop filter [9], the state-of-the-art spline interpolation-based timing estimation [10] and the joint carrier frequency offset (CFO), gridless CE, and data detection [11]. In summary, the above algorithms operate under the assumption of uniform residual Doppler frequencies across all channel taps. This channel model can be regarded as an LTI channel with a common CFO. It is expected that, for a UWA channel which exhibits double dispersive time-varying characteristic, these

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algorithms may be ineffective as the simplified model adopted by them is inconsistent with realistic channels, where the residual Doppler frequencies of the channel taps are different.

In contrast to the LTI channels with a common CFO, a more practical alternative considers each channel path to have its own distinct residual Doppler frequency. After resampling to remove the dominant wideband Doppler effect, the received signal effectively encounters a channel in which each path exhibits a different Doppler frequency. Eggen et al. [12] proposed a time-updated RLS algorithm to estimate the channel in delay-Doppler domain, with a subsequent MMSE equalizer for symbol recovery. Exploiting the inherent sparsity of UWA channels in the delay-Doppler domain, various compressive sensing-based algorithms, such as least squares matching-pursuit (LSMP) [13], orthogonal MP (OMP) [14], compressed sampling MP [15], sparsity adaptive MP [16], sparse Bayesian learning (SBL) [17] have been proposed, significantly yielding lower bit error rates (BERs) without a substantial increase in computational complexity. It is worth noting that these approaches are on-grid methods, i.e., the delays and the residual Doppler frequencies of all the paths are discretized into predefined grids, leading to model mismatch issues [18].

The developed UWA communication model is a bilinear model when the residual Doppler frequencies are known. Therefore, performing the joint CE and data decoding using the PBiGAMP is anticipated to bring performance gains. As the residual Doppler frequencies are continuous-valued parameters, gridless Doppler frequencies estimation algorithms are expected to provide more accurate channel estimates, yielding lower BERs. Motivated by these considerations, we develop an iterative receiver for joint channel taps estimation, gridless Doppler frequency estimation and symbol detection, based on PBiGAMP algorithm. The main contributions of this work can be summarized as follows: First, the input–output relationship of channel with nonuniform Doppler shifts is established, wherein the residual Doppler frequencies are modeled as continuous-valued variables. A detailed analysis reveals that if certain paths exhibit strong amplitudes, small delay difference, yet relatively large differences in Doppler shifts, the resulting leakage can be severe and diminish the model's accuracy. Otherwise the channel model closely approximates the original model. Second, by treating the Doppler frequencies as deterministic unknown parameters, assigning a Bernoulli–Gaussian prior to the channel taps and a discrete prior to the symbols, a probabilistic formulation is developed. Third, an iterative receiver is proposed from a modular perspective, comprising a decoder module and a parametric bilinear generalized approximate message passing for nonuniform Doppler shifts (PBiGAMP-NDS) module. The decoder module provides extrinsic symbol information that serves as a prior in the PBiGAMP-NDS module. In turn, PBiGAMP-NDS jointly estimates the sparse channel taps, the gridless Doppler frequencies and the symbols, the noise variance and the nuisance parameters of the prior distribution. Notably, the Newton method is incorporated to refine Doppler frequency estimates without discretization. PBiGAMP-NDS outputs extrinsic symbol information, which is then used as a pseudolikelihood in the low density parity check (LDPC) decoder. These two modules iteratively exchange extrinsic information to refine the channel estimates, the gridless Doppler frequencies and the data symbols until convergence. Finally, the performance of the proposed PBiGAMP-NDS-based iterative receiver is validated both in numerical simulations and sea trials. It is shown that in simulation scenario 2 presented in Section IV-A, for coded BER being 10^{-3}, the SNR gain of PBiGAMP-NDS is 2.5 dB, compared to the original PBiGAMP. In sea trials, its coded BER is found to be an order of magnitude lower than that of the original PBiGAMP.

The rest of this article is organized as follows. The input–output relationship of the UWA under nonuniform Doppler shifts and probabilistic formulation is given in Section II. Section III describes the proposed algorithm in details. Section IV presents simulation results and performance evaluation. Verification with sea trial data is provided in Section V. Finally, Section VI concludes this article.

Notations used throughout this article include: boldface uppercase $\mathbf{A}$ for matrices, boldface lowercase $\mathbf{a}$ for vectors, a_i for the ith element, and $z_m^{(l,j)}$ for the (m,l,j)th element of a 3-D tensor. $\mathbf{I}_M$ and $\mathbf{0}_{M,N}$ represent the $M \times M$ identity matrix and an all-zero matrix of size $M \times N$, respectively. Transpose and Hermitian operations are denoted by $(\cdot)^{\mathrm{T}}$ and $(\cdot)^{\mathrm{H}}$, while $\odot$ and $(\cdot)^{\odot 2}$ signify element-wise multiplication and square, respectively. $\mathbf{a}/\mathbf{b}$ denotes element-wise division of vector $\mathbf{a}$ and $\mathbf{b}$. The probability density function (PDF) of a multivariate complex Gaussian random vector $\mathbf{x}$ with mean $\hat{\mathbf{x}}$ and covariance $\mathbf{\Sigma}$ is represented as $\mathcal{CN}(\mathbf{x}|\hat{\mathbf{x}}, \mathbf{\Sigma})$. $\Re\{\cdot\}$ denotes the real part of a complex number, vector or matrix.

II. System Model and Probabilistic Formulation

A. System Model

In this article, we employ a zero-padding (ZP) single-carrier system, where each baseband package takes the form of

$$\mathbf{x} = [\mathbf{x}_P^{\mathrm{T}}, \mathbf{x}_D^{\mathrm{T}}, \mathbf{x}_G^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{C}^{N_c \times 1}. \tag{1}$$

Here, $\mathbf{x}_P \in \mathbb{R}^{N_p \times 1}$ is the pilot sequence, drawn from the pilot symbol alphabet $\mathcal{A}_p$, which is typically a binary phase-shift keying (BPSK) alphabet. The information sequence $\mathbf{x}_D \in \mathcal{A}_d^{N_{\mathrm{cd}} \times 1}$ is selected from the symbol alphabet $\mathcal{A}_d \triangleq \{\alpha_0, \ldots, \alpha_{2^Q-1}\}$, which is commonly a 2^Q-ary quadrature amplitude modulation or a quadrature phase-shift keying (QPSK) alphabet. In addition, $\mathbf{x}_G = \mathbf{0}_{N_g}$ is the zero guard sequence. The information sequence $\mathbf{x}_D$ carries N_b information bits $\mathbf{b} = [b_0, b_1, \ldots, b_{N_b-1}]^{\mathrm{T}} \in \{0,1\}^{N_b \times 1}$. These bits are encoded by an encoder at rate R, producing a coded bit stream of length $N_{\mathrm{code}} = N_b/R$. The coded bits are then divided into N_{cd} subgroups, i.e.,

$$\mathbf{c} = \mathcal{C}(\mathbf{b}) = [\mathbf{c}_0^{\mathrm{T}}, \mathbf{c}_1^{\mathrm{T}}, \ldots, \mathbf{c}_{N_{\mathrm{cd}}-1}^{\mathrm{T}}]^{\mathrm{T}} \in \{0,1\}^{N_{\mathrm{code}} \times 1}$$

where $N_{\mathrm{cd}} = N_{\mathrm{code}}/Q$, $\mathbf{c}_n = [c_{n,0}, c_{n,1}, \ldots, c_{n,Q-1}]^{\mathrm{T}}$, and $\mathcal{C}: \{0,1\}^{N_b \times 1} \to \{0,1\}^{N_{\mathrm{code}} \times 1}$ is the coding function. Next $\mathbf{c}$ is mapped to $\mathbf{x}_D = \mathcal{M}(\mathbf{c})$ according to the symbol alphabet, with

$$\mathcal{M}(\cdot): \{0,1\}^{N_{\mathrm{code}} \times 1} \to \{\alpha_0, \ldots, \alpha_{2^Q-1}\}^{N_{\mathrm{cd}} \times 1}$$

denoting the mapping function. The total length of the whole sequence is $N_c = N_p + N_{\mathrm{cd}} + N_g$. The indexes of pilot,

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data and zero guard part of $\mathbf{x}$ are defined as $\mathcal{P} \triangleq \{0, 1, \ldots, N_p - 1\}$, $\mathcal{D} \triangleq \{N_p, N_p + 1, \ldots, N_p + N_{\rm cd} - 1\}$ and $\mathcal{G} \triangleq \{N_p + N_{\rm cd}, N_p + N_{\rm cd} + 1, \ldots, N_c - 1\}$, respectively.

Let $\tilde{x}(t)$ be the passband continuous-time signal

$$\tilde{x}(t) = \Re\left\{\sum_{n=0}^{N_c-1} x_n g(t - nT_s) \mathrm{e}^{\mathrm{j}2\pi f_c t}\right\} \tag{2}$$

where $\{x_n\}_{n=0}^{N_c-1}$ is the discrete symbol sequence, $g(t)$ is the transmitted/received pulse-shaping filter, and f_c is the carrier frequency. Generally, the time-varying CIR of the UWA channel is represented as

$$\tilde{h}(t, \tau) = \sum_{k=0}^{p_0-1} \tilde{\eta}_k \delta(t - \tau_k(t)) \tag{3}$$

where $\tilde{\eta}_k$ and $\tau_k(t)$ denote, respectively, the overall attenuation and propagation delay at time t of path k, p_0 denotes the number of paths. Here the propagation delay $\tau_k(t)$ is modeled as [19]

$$\tau_k(t) = \tau_k - a_k t \tag{4}$$

where a_k and τ_k are, respectively, the Doppler rate and the static delay of the kth path. In addition, it is assumed that all the Doppler rates lie near the common Doppler rate a such that

$$a_k = a + \delta a_k \tag{5}$$

where δa_k can be viewed as the residual Doppler rate of the kth path and $|\delta a_k|$ is a very small value. After passing through the UWA channel (3), downconverting, resampling, and CFO compensation whose details are put in Appendix A, the received baseband signal can be expressed as

$$y(t) = \sum_{k=0}^{p_0-1} \eta_k \sum_{n=0}^{N_c-1} x_n g\left(t - \tau_k - nT_s\right) \mathrm{e}^{\mathrm{j}\nu_k t} + \widetilde{w}(t) \tag{6}$$

where the equivalent complex amplitude η_k and the equivalent residual Doppler angular frequency ν_k of the kth path are defined as

$$\eta_k \triangleq \tilde{\eta}_k \mathrm{e}^{-\mathrm{j}2\pi f_c \tau_k} \tag{7a}$$

$$\nu_k \triangleq 2\pi f_c \frac{a + \delta a_k - \hat{a}}{1 + \hat{a}} \tag{7b}$$

respectively.

The received baseband signal $y(t)$ is first passed through a pulse-shaping filter $g(t)$ and then sampled at a rate $1/T_s$ samples/s. The sampled value y_n at the nT_s instant is

$$y_n = \int y(t) g(nT_s - t) \mathrm{d}t. \tag{8}$$

Defining $p(t) = g(t) * g(t)$, the sampled signal is given by [20][1]

$$y_n = \sum_{l=0}^{L-1} \sum_{k=0}^{p_0-1} \eta_k \mathrm{e}^{\mathrm{j}\nu_k (n-l)T_s} p(lT_s - \tau_k) x_{n-l} + \breve{w}_n \tag{9}$$

where $L = [\tau_{\max}/T_s] + N_{\rm sps}$, $[\cdot]$ denotes rounded operation, and $\tau_{\max}$ is the maximum delay, i.e., $\tau_{\max} = \max(\tau_0, \tau_1, \ldots, \tau_{p_0-1})$, $N_{\rm sps}$ is the half of the effective duration normalized by T_s of $p(t)$.[2] From the sampling perspective, a channel path with delay τ_k will generate $2N_{\rm sps} - 1$ channel taps at sample indexes

$$\mathcal{L}_k \triangleq \begin{cases} \{-N_{\rm sps} + l_k + 1, \ldots, l_k, N_{\rm sps} + l_k\}, & \text{if } \tau_k > l_k T_s \\ \{-N_{\rm sps} + l_k, \ldots, l_k, N_{\rm sps} + l_k - 1\}, & \text{if } \tau_k < l_k T_s \\ \{l_k\}, & \text{if } \tau_k = l_k T_s \end{cases} \tag{10}$$

where $l_k = [\tau_k/T_s]$. For each $l \in [0, L-1]$, define the path set $\mathcal{K}_l = \{k | l \in \mathcal{L}_k\}$. In this set, the path with the most significant energy at l is denoted as the path k^l, i.e.,

$$k^l = \underset{k \in \mathcal{K}_l}{\operatorname{argmax}} |\eta_k p(lT_s - \tau_k)|$$

and k^l is dependent only on l. For other paths in $\mathcal{K}_l$, they can be categorized into two types. For the first type, their Doppler frequencies are close to that of path k^l. While for the second type, their Doppler frequencies are somewhat different from that of path k^l. We combine the first type of paths and path k^l into a set $\tilde{\mathcal{K}}_l$, i.e., we have $|\nu_k - \nu_{k^l}| < \nu_{\rm th}, \forall k \in \tilde{\mathcal{K}}_l$. Hence, the sampled signal can be written as

$$y_n = \underbrace{\sum_{l=0}^{L-1} \sum_{k \in \tilde{\mathcal{K}}_l} \eta_k \mathrm{e}^{\mathrm{j}\nu_k ((n-l)T_s)} p(lT_s - \tau_k) x_{n-l}}_{\text{the first term}} + \underbrace{\sum_{l=0}^{L-1} \sum_{k \in \mathcal{K}_l \backslash \tilde{\mathcal{K}}_l} \eta_k \mathrm{e}^{\mathrm{j}\nu_k ((n-l)T_s)} p(lT_s - \tau_k) x_{n-l}}_{\text{the second term}} + \breve{w}_n \tag{11}$$

where the second term is interpreted as the leakage component denoted as $w_n^{\rm lk}$. Equation (11) can be reformulated as

$$y_n = \sum_{l=0}^{L-1} \sum_{k \in \tilde{\mathcal{K}}_l} \eta_k \mathrm{e}^{-\mathrm{j}\nu_k l T_s} p(lT_s - \tau_k) \mathrm{e}^{\mathrm{j}\nu_k n T_s} x_{n-l} + w_n^{\rm lk} + \tilde{w}_n. \tag{12}$$

By definition, if $|\tilde{\mathcal{K}}_l| > 1, \forall k \in \tilde{\mathcal{K}}_l$, the Doppler frequencies ν_k are close to a common frequency ν_{k^l}. For a fixed l, the inner summation of the first term of (12) is approximated as

$$\begin{aligned} &\sum_{k \in \tilde{\mathcal{K}}_l} \eta_k \mathrm{e}^{-\mathrm{j}\nu_k l T_s} p(lT_s - \tau_k) \mathrm{e}^{\mathrm{j}\nu_k n T_s} x_{n-l} \\ &\quad \approx \mathrm{e}^{\mathrm{j}\nu_{k^l} n T_s} x_{n-l} \sum_{k \in \tilde{\mathcal{K}}_l} \eta_k \mathrm{e}^{-\mathrm{j}\nu_{k^l} l T_s} p(lT_s - \tau_k). \end{aligned} \tag{13}$$

Let $w_{n,l}^{\nu}$ be the approximation error for a given $l \in [0, L-1]$ in (13) and define the equivalent noise $w_n \triangleq \tilde{w}_n + \sum_l w_{n,l}^{\nu} + w_n^{\rm lk}$, which is assumed to be independent identically distributed (i.i.d.) and $w_n \sim \mathcal{CN}(w_n | 0, \sigma^2)$. Given n, the right-hand side of (13) can be determined by l. Therefore, by introducing

$$u_l \triangleq \sum_{k \in \tilde{\mathcal{K}}_l} \eta_k \mathrm{e}^{-\mathrm{j}\nu_{k^l} l T_s} p(lT_s - \tau_k) \tag{14a}$$

$$\theta_l \triangleq \nu_{k^l} T_s. \tag{14b}$$

[1] Beygi et al. [20] give another reformulation and proposes the parametric low rank atomic norm (PLAN) method. Their model does not suffer from the leakage issues, but the Doppler frequencies can not be estimated accurately as they are too close and the length of the pilots is short in this setting.

[2] i.e., the support of $p(t)$ is $[-N_{\rm sps}T_s, N_{\rm sps}T_s]$, and a reasonable choice for $N_{\rm sps}$ is $N_{\rm sps} = 4$ [21].

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Equation (12) can be reformulated as

$$y_n = \sum_{l=0}^{L-1} u_l e^{j\theta_l n} x_{n-l} + w_n. \tag{15}$$

From the above analysis, the simplified model (15) closely approximates the original model (9) whenever, $\forall n$ both the leakage component w_n^{lk} (12) and the approximation error $\sum_l w_{n,l}^{\nu}$ (13) remain small. This condition is satisfied if 1) path delays and Doppler shifts differ only slightly, making $|\mathcal{K}_l \backslash \tilde{\mathcal{K}}_l|$ small, $\forall l$, or 2) the path delays are large enough such that $\forall k \in \mathcal{K}_l \backslash \tilde{\mathcal{K}}_l$, $|p(lT_s - \tau_k)|$ is negligible. Later we reveal how the paths' delays and Doppler frequencies affects the accuracy of the model (15) through numerical experiments. A very typical example is that all the residual Doppler shifts are the same, i.e., $\nu_k = \nu$ including the LTI channel model as a special case, where $\nu_k = \nu = 0$, $\forall k$. In this case $\mathcal{K}_l \backslash \tilde{\mathcal{K}}_l = \emptyset$ (11), both the leakage component w_n^{lk} (12) and the approximation error $\sum_l w_{n,l}^{\nu}$ (13) are 0, model (15) is exact and can be simplified as

$$y_n = e^{j\theta n} \sum_{l=0}^{L-1} u_l x_{n-l} + w_n. \tag{16}$$

As shown in Appendix C, a brief implementation of the PBiGAMP-based algorithm, namely PBiGAMP-CFO to solve the joint CE, data decoding and the CFO estimation problem (16) is given.

B. Probabilistic Formulation

The above scalar model (15) can also be recast in matrix vector form. Stacking N_c successively received samples together and assuming that both u_l and θ_l remain constant over one ZP data package yields

$$\mathbf{y} = \sum_{l=0}^{L-1} u_l \mathbf{Z}^{(l)}(\theta_l)\mathbf{x} + \mathbf{w} \tag{17}$$

where $\mathbf{y} \triangleq [y_0, \ldots, y_{N_c-1}]^{\mathrm{T}}$, $\mathbf{w} \triangleq [w_0, \ldots, w_{N_c-1}]^{\mathrm{T}}$, and

$$\mathbf{Z}^{(l)}(\theta_l) = \begin{bmatrix} \mathbf{0}_{l\times(N_c-l)} & \mathbf{0}_{l\times l} \\ \operatorname{diag}(\mathbf{a}_{l,N_c}(\theta_l)) & \mathbf{0}_{(N_c-l)\times l} \end{bmatrix} \tag{18}$$

with $\mathbf{a}_{l,N_c}(\theta) \triangleq [e^{j\theta l}, \ldots, e^{j\theta(N_c-1)}]^{\mathrm{T}} \in \mathbb{C}^{(N_c-l)\times 1}$. Notably (17) generalizes two particular models. The first is that each channel tap has an identical equivalent Doppler frequency θ, i.e., $\theta_l = \theta\ \forall l$. In this setting, by defining

$$\mathbf{Z}(\theta) \triangleq \operatorname{diag}(\mathbf{a}_{0,N_c}(\theta)) \in \mathbb{C}^{N_c\times N_c}$$

(16) reduces to

$$\mathbf{y} = \mathbf{Z}(\theta)\mathbf{U}\mathbf{x} + \mathbf{w} \tag{19}$$

where $\mathbf{U} \in \mathbb{C}^{N_c\times N_c}$ is a Toeplitz circulant matrix whose first column is $[u_0, \ldots, u_{L-1}, \mathbf{0}_{1\times(N_c-L)}]^{\mathrm{T}}$. In UWA systems, θ in (19) is commonly referred to as the CFO [2], [11]. A further simplified case sets $\theta = 0$ meaning that the Doppler effect is fully removed by resampling and CFO compensation, which recovers the standard LTI channel model. Here, we model $\mathbf{u} \triangleq [u_0, \ldots, u_{L-1}]^{\mathrm{T}}$ and $\mathbf{x}$ as two random vectors. Given the sparsity typically found in UWA channels, a separable Bernoulli–Gaussian prior is imposed on $\mathbf{u}$, i.e.,

$$p(\mathbf{u}) = \prod_{l=0}^{L-1} p(u_l; \lambda, \zeta, \psi) \tag{20}$$

where

$$p(u_l; \lambda, \zeta, \psi) = (1-\lambda)\delta(u_l) + \lambda \mathcal{CN}(u_l; \zeta, \psi). \tag{21}$$

Here, λ is the probability that the equivalent complex amplitude of the lth path u_l is active, i.e., $u_l \neq 0$, ζ and ψ are the mean and variance of the equivalent complex amplitude of the lth path u_l. The prior of $\mathbf{x}$ can be expressed as

$$p(\mathbf{x}) = p(\mathbf{x}_P)p(\mathbf{x}_D)p(\mathbf{x}_G) \tag{22}$$

where $p(\mathbf{x}_G) = \delta(\mathbf{x}_G)$, $p(\mathbf{x}_P) = \delta(\mathbf{x}_P - \mathbf{x}_{P_0})$, and $\mathbf{x}_{P_0}$ is the known pilot, $p(\mathbf{x}_D)$ can be obtained in the following way. The raw information bit is drawn from an equiprobable binary distribution, therefore the probability mass function (PMF) of $p(\mathbf{b})$ is

$$p(\mathbf{b}) = \prod_{k=0}^{N_b-1} p(b_k) = \prod_{k=0}^{N_b-1} (0.5 \times 1_{[b_k\in\{0,1\}]}) \tag{23}$$

where $1_{[\cdot]}$ denotes the indicator function. After coding and mapping to symbol alphabet, the PMF of coded bits $\mathbf{c}$ conditioned on $\mathbf{b}$ and mapped symbols conditioned on $\mathbf{c}$ are expressed as $p(\mathbf{c}|\mathbf{b}) = 1_{[\mathbf{c}=\mathcal{C}(\mathbf{b})]}$ and $p(\mathbf{x}_D|\mathbf{c}) = 1_{[\mathbf{x}_D=\mathcal{M}(\mathbf{c})]}$, respectively. Thus, $p(\mathbf{x}_D)$ can be calculated by integrating the joint PMF $p(\mathbf{x}_D, \mathbf{c}, \mathbf{b})$ with respect to $\mathbf{c}$ and $\mathbf{b}$, i.e.,

$$p(\mathbf{x}_D) = \sum_{\mathbf{b}} \sum_{\mathbf{c}} p(\mathbf{b})p(\mathbf{c}|\mathbf{b})p(\mathbf{x}_D|\mathbf{c}). \tag{24}$$

As for the equivalent Doppler frequencies $\boldsymbol{\theta} \triangleq [\theta_0, \theta_1, \ldots, \theta_{L-1}]^{\mathrm{T}}$, we model them as deterministic unknown parameters and $\boldsymbol{\theta} \in [-\nu_{\max}T_s, \nu_{\max}T_s]^{L\times 1}$.

According to the measurement model (17), the likelihood function is

$$p(\mathbf{y}|\mathbf{x}, \mathbf{u}; \boldsymbol{\theta}, \sigma^2) = \mathcal{CN}\left(\mathbf{y} \middle| \sum_{l=0}^{L-1} u_l \mathbf{Z}^{(l)}(\theta_l)\mathbf{x}, \sigma^2 \mathbf{I}_N\right) \tag{25}$$

which is parameterized by the two unknown parameters $\boldsymbol{\theta}$ and σ^2. A standard strategy is to adopt the type II maximum likelihood (ML) method [22] or "Evidence procedure" [23]. First, the deterministic unknown parameters $\boldsymbol{\theta} \in [-\nu_{\max}T_s, \nu_{\max}T_s]^{L\times 1}$, σ^2, λ, ζ, and ψ are estimated via maximizing the marginal likelihood function $p(\mathbf{y}; \boldsymbol{\theta}, \sigma^2, \lambda, \zeta, \psi)$, i.e.,

$$\begin{aligned}(\boldsymbol{\theta}_{\text{ML}}, \lambda_{\text{ML}}, \zeta_{\text{ML}}, \psi_{\text{ML}}, \sigma^2_{\text{ML}}) &= \underset{\boldsymbol{\theta},\lambda,\zeta,\psi,\sigma^2}{\operatorname{argmax}}\ p(\mathbf{y}; \boldsymbol{\theta}, \lambda, \zeta, \psi, \sigma^2) \\ &= \underset{\boldsymbol{\theta},\lambda,\zeta,\psi,\sigma^2}{\operatorname{argmax}} \sum_{\mathbf{x}} \int p(\mathbf{y}|\mathbf{x}, \mathbf{u}; \boldsymbol{\theta}, \sigma^2)p(\mathbf{x})p(\mathbf{u})\mathrm{d}\mathbf{u}\end{aligned} \tag{26}$$

where $p(\mathbf{x})$ can be obtained by substituting (24) into (22). Second, the maximum a posteriori estimate $\hat{b}_i$ of b_i and the MMSE estimate $\hat{\mathbf{u}}$ of $\mathbf{u}$ are

$$\hat{b}_i = \underset{b_i\in\{0,1\}}{\operatorname{argmax}}\, p(b_i|\mathbf{y}) \tag{27}$$

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and

$$\hat{\mathbf{u}} = \mathrm{E}_{\mathbf{u}\sim p(\mathbf{u}|\mathbf{y})}[\mathbf{u}] \tag{28}$$

where the posterior PMF of b_i and the posterior PDF of $\mathbf{u}$ are

$$\begin{aligned} p(b_i|\mathbf{y}) \propto & \sum_{\mathbf{b}\setminus b_i,\mathbf{x},\mathbf{c}} \int_{\mathbf{u}} p(\mathbf{y}|\mathbf{u},\mathbf{x};\boldsymbol{\theta}_{\mathrm{ML}};\sigma^2_{\mathrm{ML}}) \\ & \times p(\mathbf{u};\lambda_{\mathrm{ML}},\zeta_{\mathrm{ML}},\psi_{\mathrm{ML}})p(\mathbf{x}|\mathbf{c})p(\mathbf{c}|\mathbf{b})p(\mathbf{b})\mathrm{d}\mathbf{u} \\ = & \sum_{\mathbf{b}\setminus b_i,\mathbf{c}} p(\mathbf{c}|\mathbf{b})p(\mathbf{b}) \sum_{\mathbf{x}} \int_{\mathbf{u}} p(\mathbf{y}|\mathbf{u},\mathbf{x};\boldsymbol{\theta}_{\mathrm{ML}};\sigma^2_{\mathrm{ML}}) \\ & \times \prod_{l=0}^{L-1} p(u_l;\lambda_{\mathrm{ML}},\zeta_{\mathrm{ML}},\psi_{\mathrm{ML}}) \\ & \times \prod_{j\in\mathcal{D}} p(x_j|\mathbf{c}_{j-N_p})\mathrm{d}\mathbf{u} \end{aligned} \tag{29}$$

and

$$\begin{aligned} p(\mathbf{u}|\mathbf{y}) \propto & \sum_{\mathbf{b},\mathbf{c},\mathbf{x}} p(\mathbf{y}|\mathbf{u},\mathbf{x};\boldsymbol{\theta}_{\mathrm{ML}};\sigma^2_{\mathrm{ML}}) \\ & \times p(\mathbf{u};\lambda_{\mathrm{ML}},\zeta_{\mathrm{ML}},\psi_{\mathrm{ML}})p(\mathbf{x}|\mathbf{c})p(\mathbf{c}|\mathbf{b})p(\mathbf{b}) \\ = & \sum_{\mathbf{b},\mathbf{c},\mathbf{x}} p(\mathbf{c}|\mathbf{b})p(\mathbf{b})p(\mathbf{y}|\mathbf{u},\mathbf{x};\boldsymbol{\theta}_{\mathrm{ML}};\sigma^2_{\mathrm{ML}}) \prod_{l=0}^{L-1} \\ & \times p(u_l;\lambda_{\mathrm{ML}},\zeta_{\mathrm{ML}},\psi_{\mathrm{ML}}) \prod_{j\in\mathcal{D}} p(x_j|\mathbf{c}_{j-N_p}) \end{aligned} \tag{30}$$

respectively. All the optimization problems (26)–(28) are computationally intractable. Therefore, in the ensuing section, we resort to the approximate Bayesian method.

III. Joint Doppler, CE, and Data Detection

To address (26) in a computationally efficient manner, we exploit the problem's structure. Here, we introduce a hidden variable

$$\mathbf{z} \triangleq \mathbf{z}(\mathbf{u},\mathbf{x};\boldsymbol{\theta}) \triangleq \sum_{l=0}^{L-1} u_l \mathbf{Z}^{(l)}(\theta_l)\mathbf{x} \tag{31}$$

which is parameterized by two random variables $\mathbf{u}$, $\mathbf{x}$, and one deterministic parameter $\boldsymbol{\theta}$. By investigating the measurement model (17), we have the following observations.

1) When $\boldsymbol{\theta}$ is fixed, $\mathbf{z}$ is bilinear with respect to $\mathbf{x}$ and $\mathbf{u}$, which means when either $\mathbf{x}$ or $\mathbf{u}$ is fixed, $\mathbf{z}$ is linear with respect to the other variable.
2) The conditional likelihood $p(\mathbf{y}|\mathbf{z})$ is separable, i.e., $p(\mathbf{y}|\mathbf{z}) = \prod_{n=0}^{N_c-1} p(y_n|z_n)$.
3) If $\mathbf{x}$ and $\mathbf{u}$ are both fixed, the hidden variable $\mathbf{z}$ is nonlinear with respect to $\boldsymbol{\theta}$.

Points 1) and 2) imply that when $\boldsymbol{\theta}$ is known, the problem can be solved efficiently by PBiGAMP algorithm provided that both the priors of $\mathbf{x}$ and $\mathbf{u}$ are separable, which is not the case as the prior of $\mathbf{x}$ is nonseparable, while point 3) indicates that we can adopt the type II ML method [22] to estimate $\boldsymbol{\theta}$ by maximizing $p(\mathbf{y};\boldsymbol{\theta})$, which is the PDF via marginalizing the joint approximated PDF $p(\mathbf{y},\mathbf{x},\mathbf{u};\boldsymbol{\theta})$ with respect to random variables $\mathbf{x}$ and $\mathbf{u}$. To reduce the computational complexity, we adopt the block coordinate descent method [24] to refine all the equivalent Doppler frequencies, and apply Newton's method to refine each one individually.

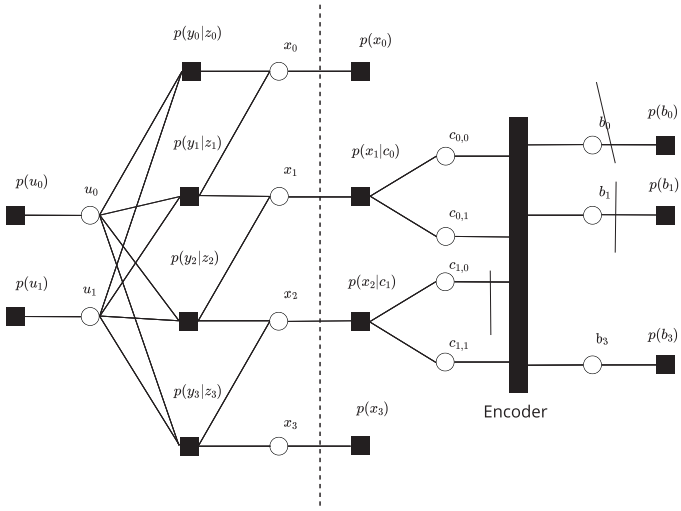


Fig. 1. Example factor graph describing the single-carrier system over narrowband time-varying channel, when $N_p = 1, N_{\mathrm{cd}} = 2, N_g = 1, Q = 2, L = 2$. Here, the white circle denotes the variable node and the square black denotes the factor node.

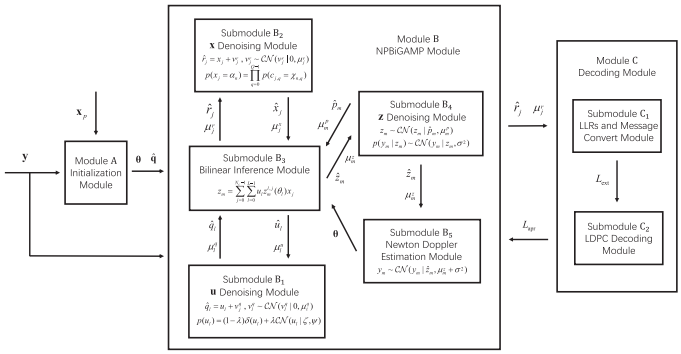


Fig. 2. Proposed modularized iterative receiver. Three modules, named initialization module, decoding module, and the PBiGAMP-NDS module, the latter two modules iteratively exchange the extrinsic information to perform the joint CE, gridless Doppler frequency estimation and data detection.

Having established the probabilistic formulation in Section II-B, the factor graph depicted in Fig. 1 can be obtained. Generally, the elements of $\mathbf{x}$ are correlated, and the prior of $\mathbf{x}$ is nonseparable. Therefore, we partition the factor graph into two subgraphs, separated by a dashed line. For the right and left subgraph, we introduce a standard decoder module and the PBiGAMP-NDS module, respectively. The decoder module outputs the extrinsic information of the symbol and feeds it into the PBiGAMP-NDS module. Since the extrinsic information of the symbol feeding into the left factor subgraph is separable, efficient estimation of $\mathbf{x}$, $\mathbf{u}$, and $\boldsymbol{\theta}$ through the proposed PBiGAMP-NDS is possible. As for the PBiGAMP-NDS module shown in Fig. 2, it is delineated into the following steps.

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1) Denoising of $\mathbf{u}$ and $\mathbf{x}$ in submodules B_1 and B_2, respectively.
2) Calculating extrinsic mean and variance of $\mathbf{z}$ in submodules B_3.
3) Denoising of $\mathbf{z}$ in submodules B_4.
4) Refining $\boldsymbol{\theta}$ to update the matrix $\mathbf{Z}^{(l)}(\theta_l)$ by Newton method in submodules B_5.
5) Calculating the extrinsic mean and variance of $\mathbf{x}$ and $\mathbf{u}$ in submodules B_3.

After several iterations, the PBiGAMP-NDS module provides the extrinsic mean and variance of $\mathbf{x}$, which are fed into the decoder module as the likelihood of the symbol. The decoder module converts the likelihood of the symbol to extrinsic log-likelihood ratios (LLRs) of the bits in submodule C_1, denoted as L_{ext}. The LDPC decoder in submodule C_2 then performs a standard decoding algorithm to calculate the posterior L_{pos}. We then obtain the new prior LLR L_{apr} by $L_{\text{apr}} = L_{\text{pos}} - L_{\text{ext}}$, which will later be utilized as the prior of the symbol in Step 1), thereby closing the loop of the proposed iterative receiver. Additionally, between every loop, an expectation–maximization (EM) algorithm is executed to refine parameters σ^2, λ, ζ, and ψ. We now detail the steps of the proposed PBiGAMP-NDS-based iterative receiver, whose structure is depicted in Fig. 2.

A. Parametric Bilinear Generalized Approximate Message Passing for Nonuniform Doppler Shifts

The PBiGAMP-NDS consists of initialization and several submodules. For clarity, the details of the denoising submodules are put in Appendix B.

1) Initialization: At first, we initialize the extrinsic mean and variance of $\mathbf{x}$ and $\mathbf{u}$, along with the Doppler frequencies $\boldsymbol{\theta}$, respectively. For $\mathbf{x}$, a straightforward choice is $\mathbf{r} = [\mathbf{x}_{\mathcal{P}}^{\mathrm{T}}, \mathbf{0}_{1\times N_{cd}}, \mathbf{x}_{\mathcal{G}}^{\mathrm{T}}]^{\mathrm{T}}$, $\boldsymbol{\mu}^r = [\epsilon\mathbf{1}_{1\times N_p}, \mathbf{1}_{1\times N_{\text{cd}}}, \epsilon\mathbf{1}_{1\times N_g}]^{\mathrm{T}}$ where ϵ is a small value such as 10^{-4}. For $\mathbf{u}$, we have its extrinsic variance $\boldsymbol{\mu}^q = \mathbf{1}_{L\times 1}$. To initialize the extrinsic mean of $\mathbf{u}$, i.e., $\hat{\mathbf{q}}$ and $\boldsymbol{\theta}$, a greedy approach is employed. By treating $\boldsymbol{\theta} = [\theta_0, \theta_1, \ldots, \theta_{L-1}]^{\mathrm{T}}$ and $\mathbf{u} = [u_0, u_1, \ldots, u_{L-1}]^{\mathrm{T}}$ as unknown deterministic parameters, the log-likelihood function $\log p(\mathbf{y}_p; \boldsymbol{\theta}, \mathbf{u})$ can be expressed as

$$\begin{aligned}\log p(\mathbf{y}_p;\boldsymbol{\theta},\mathbf{u}) &= \text{const} \\ &- \left(\mathbf{y}_p - \sum_{l'\in\mathcal{L}\setminus l} u_{l'}\mathbf{Z}_p^{(l')}(\theta_{l'})\mathbf{x}_p - u_l\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\right)^{\mathrm{H}} \\ &\left(\mathbf{y}_p - \sum_{l'\in\mathcal{L}\setminus l} u_{l'}\mathbf{Z}_p^{(l')}(\theta_{l'})\mathbf{x}_p - u_l\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\right)/\sigma^2 \end{aligned} \tag{32}$$

where const denotes the irrelevant terms independent of $\boldsymbol{\theta}$ and $\mathbf{u}$, $\mathcal{L} \triangleq \{0, 1, \ldots, L-1\}$. We employ the greedy algorithm similar to the Newtonized orthogonal matching pursuit (NOMP) [25] to obtain the initial Doppler and the amplitude estimates. First, a single virtual path corresponding to $L = 1$ is addressed and then the multiple virtual paths are handled. For a single virtual path, we have the measurement model

$$\mathbf{y}_p = u_l\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p + \mathbf{w}. \tag{33}$$

Consequently, the ML estimation of the path index l, the amplitude u_l and the Doppler frequency l can be equivalently formulated as

$$\underset{l,\theta_l,u_l}{\text{minimize}} \ \|\mathbf{y}_p - u_l\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\|_2^2 \triangleq S(l,\theta_l,u_l). \tag{34}$$

Given l and θ_l being fixed, $S(l,\theta_l,u_l)$ is a quadratic function of u_l. The optimal $u_l^\star$ is

$$u_l^\star = \frac{\left(\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\right)^{\mathrm{H}}\mathbf{y}_p}{\|\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\|_2^2}. \tag{35}$$

Substituting $u_l^\star$ (35) in $S(l,\theta_l,u_l)$ (34), the optimization problem (34) can be simplified to be

$$\underset{l,\theta_l}{\text{maximize}} \ \frac{\left|\left(\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\right)^{\mathrm{H}}\mathbf{y}_p\right|^2}{\|\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p\|_2^2} \triangleq G(l,\theta_l). \tag{36}$$

The above problem is solved by discretizing θ_l into a finite number of grids predefined as Φ_g. For each grid $\theta_g \in \Phi_g$ and each virtual path l, we evaluate $G(l,\theta_g)$. The index $\hat{l}$ and the coarse estimate $\hat{\theta}_g$ yielding the maximum of $G(l,\theta_g)$ are chosen as the coarse estimates. Then, we use the Newton method to refine $\hat{\theta}_g$ and use the least squares to obtain the coarse estimate $\tilde{u}$ using (35). Now we remove the effects caused by the first virtual path by defining

$$\mathbf{y}_p^{\text{res}} = \mathbf{y}_p - \tilde{u}_{\hat{l}}\mathbf{Z}_p^{(\hat{l})}(\hat{\theta}_g)\mathbf{x}_p. \tag{37}$$

We use the residue measurements $\mathbf{y}_p^{\text{res}}$ to obtain the parameters of another virtual path similar to the previous steps. We stop when the parameters of a predefined number of virtual paths are obtained. After that, we adopt cyclic refinement similar to that of NOMP [25] to refine $\hat{\boldsymbol{\theta}}$ and $\tilde{\mathbf{u}}$. For further implementation details, refer to [25]. Details of the initialization is summarized as Algorithm 1, and we initialize $\hat{\mathbf{q}}$ by $\hat{\mathbf{q}} = \tilde{\mathbf{u}}$.

2) Denoising of $\mathbf{u}$ and $\mathbf{x}$ in Submodule B_1 and Submodule B_2, Respectively: This step involves the denoising of $\mathbf{u}$ and $\mathbf{x}$. The bilinear inference module B_3 outputs the decoupled extrinsic mean and variance of u_l and x_j from the previous iteration, denoted as $(\hat{q}_l, \mu_l^q)$ and $(\hat{r}_j, \mu_j^r)$, respectively. These values are referred to as "extrinsic" since they represent messages passed from the observation side, and do not contain immediate prior information of u_l and x_j [26]. These extrinsic values are subsequently fed into the $\mathbf{u}$ and $\mathbf{x}$ denoising module B_1 and B_2 to perform denoising. Here, denoising can be interpreted as calculating the posterior mean and variance of the corresponding variable.

3) Calculating the Extrinsic mean and Variance of $\mathbf{z}$ in Submodule B_3: In submodule B_3, the extrinsic mean and variance of $\mathbf{z}$ are calculated, which will then be fed into the submodule B_4 as the prior of $\mathbf{z}$. As shown in PBiGAMP, the extrinsic mean $\hat{p}_m$ and variance μ_m^p of $\mathbf{z}$ are [7]

$$\hat{p}_m = \hat{z}_m^{(*,*)}(\boldsymbol{\theta}) - \hat{s}_m\bar{\mu}_m^p \tag{38a}$$

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Algorithm 1: Initialization for PBiGAMP-NDS.

Input: $\mathbf{y}_p$, $\mathbf{x}_p$
Output: $\hat{\boldsymbol{\theta}}$, $\tilde{\mathbf{u}}$
1: **Initialize:** $\mathrm{it}_o = 0$, $\mathcal{L} = \{0, \ldots, L-1\}$, $\mathcal{S} = \varnothing$, $\hat{\boldsymbol{\theta}} = \mathbf{0}_{L\times 1}$, $\tilde{\mathbf{u}} = \mathbf{0}_{L\times 1}$, $\mathbf{y}_p^{\mathrm{res}} = \mathbf{y}_p$
2: **while** $\mathrm{it}_o < L$ **do**
3: $(\hat{l}, \hat{\theta}_g) \leftarrow \underset{(l\in\mathcal{L}, \theta_g\in\Phi_g)}{\operatorname{argmax}} \dfrac{\left|(\mathbf{Z}_p^{(l)}(\theta_g)\mathbf{x}_p)^{\mathrm{H}}\mathbf{y}_p^{\mathrm{res}}\right|^2}{\|\mathbf{Z}_p^{(l)}(\theta_g)\mathbf{x}_p\|_2^2}$
4: $\mathcal{S} \leftarrow \mathcal{S} \cup \hat{l}$, $[\hat{\boldsymbol{\theta}}]_{\hat{l}} = \hat{\theta}_g$
5: $\mathcal{L} \leftarrow \mathcal{L}\backslash\hat{l}$
6: $\mathbf{y}_p^{\mathrm{res}} \leftarrow \mathbf{y}_p^{\mathrm{res}}(\mathcal{S})$, where $\mathbf{y}_p^{\mathrm{res}}(\mathcal{S}) = \mathbf{y}_p - \sum_{l\in\mathcal{S}} \tilde{u}_l \mathbf{Z}_p^{(l)}(\hat{\theta}_l)$
7: update all channel gains via least squares: $[\tilde{\mathbf{u}}]_{\mathcal{S}} = \mathbf{X}^{\dagger}\mathbf{y}_p$, where $\mathbf{X}$ is a $N_p \times |\mathcal{S}|$ matrix, its column is given by $\mathbf{Z}_p^{(l)}(\theta_l)\mathbf{x}_p$ where $l \in \mathcal{S}$
8: $\mathrm{it}_o \leftarrow \mathrm{it}_o + 1$
9: **end while**

$$\mu_m^p = \bar{\mu}_m^p + \sum_{l=0}^{L-1}\sum_{j=0}^{N_c-1} \mu_l^u \mu_j^x \left|z_m^{(l,j)}(\theta_l)\right|^2 \tag{38b}$$

where

$$\bar{\mu}_m^p \triangleq \sum_{j=0}^{N_c-1} \mu_j^x \left|\hat{z}_m^{(*,j)}(\theta_l)\right|^2 + \sum_{l=0}^{L-1} \mu_l^u \left|\hat{z}_m^{(l,*)}(\theta_l)\right|^2 \tag{39a}$$

$$\hat{z}_m^{(*,*)}(\boldsymbol{\theta}) \triangleq \sum_{l=0}^{L-1}\sum_{j=0}^{N_c-1} \hat{u}_l \hat{x}_j z_m^{(l,j)}(\theta_l) \tag{39b}$$

$$\hat{z}_m^{(l,*)}(\theta_l) \triangleq \sum_{j=0}^{N_c-1} \hat{x}_j z_m^{(l,j)}(\theta_l) \tag{39c}$$

$$\hat{z}_m^{(*,j)}(\theta_l) \triangleq \sum_{l=0}^{L-1} \hat{u}_l z_m^{(l,j)}(\theta_l) \tag{39d}$$

and $\hat{s}_m$ is the nonlinear output calculated in the last iteration in Step 5).

4) Denoising of $\mathbf{z}$ *in Submodule* B_4*:* The submodule B_3 outputs the extrinsic mean $\hat{p}_m$ and variance μ_m^p of z_m in (38), which are fed into the submodule B_4 to obtain posterior mean $\hat{z}_m$ and variance ν_m^z.

5) Refining $\boldsymbol{\theta}$ *to Update the Matrix* $\mathbf{Z}^{(l)}(\theta_l)$ *in Submodule* B_5*:* In this step, the posterior means $\hat{\mathbf{z}}$ and variances $\boldsymbol{\mu}^z$ output by the submodule B_4 are fed into the Newton Doppler estimation submodule B_5 to refine $\boldsymbol{\theta}$. We postulate that $\mathbf{z}$ follows Gaussian distribution with diagonal covariance matrix, i.e., $\mathbf{z} \sim \mathcal{CN}(\mathbf{z}|\hat{\mathbf{z}}, \mathrm{diag}(\boldsymbol{\mu}^z))$. According to the measurement model (17) and suppose that the noise $\mathbf{w}$ is independent of $\mathbf{z}$, it follows that

$$\mathbf{y} \sim \mathcal{CN}\left(\mathbf{y}|\hat{\mathbf{z}}, \mathrm{diag}(\boldsymbol{\mu}^z) + \sigma^2\mathbf{I}\right). \tag{40}$$

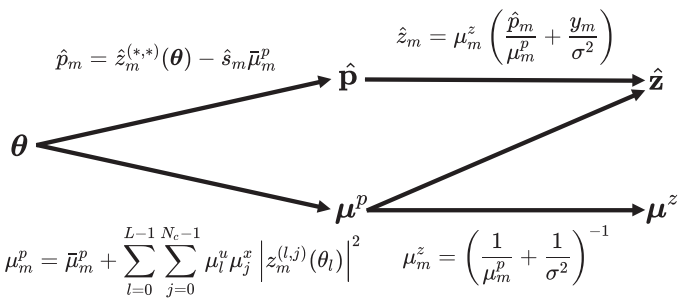


Fig. 3. Mappings $\boldsymbol{\theta} \to \hat{\mathbf{z}}$ and $\boldsymbol{\theta} \to \boldsymbol{\mu}^z$.

To estimate θ_l for all taps, we employ the type II approach [23]. In addition, we treat both the posterior means $\hat{\mathbf{z}}$ and variances $\boldsymbol{\mu}^z$ as a function of $\boldsymbol{\theta}$. Employing the type II method, the estimation problem can be formulated as

$$\underset{\boldsymbol{\theta}}{\mathrm{minimize}}\, (\mathbf{y}-\hat{\mathbf{z}})^{\mathrm{H}} \left(\mathrm{diag}(\boldsymbol{\mu}^z) + \sigma^2\mathbf{I}\right)^{-1} (\mathbf{y}-\hat{\mathbf{z}}) \triangleq g(\boldsymbol{\theta}). \tag{41}$$

Define $\hat{\theta}_l'$ as the estimate of θ_l in the last iteration. We apply Newton's method to refine the equivalent Doppler frequencies $\boldsymbol{\theta}$ cyclically to obtain $\hat{\theta}_l$, i.e., refining each equivalent Doppler frequency $\hat{\theta}_l'$ via

$$\hat{\theta}_l = \hat{\theta}_l' - \frac{\partial g(\boldsymbol{\theta})}{\partial\theta_l} \Big/ \left.\frac{\partial^2 g(\boldsymbol{\theta})}{\partial^2\theta_l}\right|_{\theta_l=\hat{\theta}_l'}, \qquad \{l|\pi_l > \lambda_{\mathrm{thre}}\} \tag{42}$$

where λ_{thre} is a manually set threshold. To calculate $\partial g(\boldsymbol{\theta})/\partial\theta_l$ and $\partial^2 g(\boldsymbol{\theta})/\partial^2\theta_l$, we first clarify how $\boldsymbol{\theta}$ is related to the posterior means $\hat{\mathbf{z}}$ and variances $\boldsymbol{\mu}^z$. By tracing the iteration and assuming that the immediate parameters, such as $\hat{\mathbf{s}}$, σ^2, and so on calculated by the last iteration do not depend on $\boldsymbol{\theta}$, the mappings of $\boldsymbol{\theta}$ to the posterior means $\hat{\mathbf{z}}$ and variances $\boldsymbol{\mu}^z$ can be shown in Fig. 3.

A closer look to the mapping of $\boldsymbol{\theta}$ to $\boldsymbol{\mu}^z$ reveals that $\boldsymbol{\mu}^z$ does not depend on $\boldsymbol{\theta}$. Specifically, $z_m^{(l,j)}(\theta_l)$ is nonzero only when $m = j + l$, i.e.,

$$z_m^{(l,j)}(\theta_l) \triangleq \begin{cases} \mathrm{e}^{\mathrm{j}\theta_l \mathrm{m}}, & m = l + j \\ 0, & m \neq l + j. \end{cases} \tag{43}$$

Therefore, μ_m^p (44) can be simplified as

$$\mu_m^p = \bar{\mu}_m^p + \sum_{j=0}^{N_c-1} \mu_j^x \sum_{l=0}^{L-1} |\hat{u}_l|^2 |z_m^{(l,j)}(\theta_l)|^2 + \sum_{l=0}^{L-1} \mu_l^u \sum_{j=0}^{N_c-1} |\hat{x}_j|^2 |z_m^{(l,j)}(\theta_l)|^2. \tag{44}$$

According to (43) and (44), the posterior variance μ_m^p does not depend on $\boldsymbol{\theta}$, yielding the first and second derivatives of μ_m^p with respect to θ_l being zero. Hence, the first and second derivatives of $g(\boldsymbol{\theta})$ with respect to θ_l are

$$\frac{\partial g(\boldsymbol{\theta})}{\partial\theta_l} = -2\Re\left\{\left(\frac{\partial\hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial\theta_l}\right)^{\mathrm{H}} \left(\mathrm{diag}(\boldsymbol{\mu}^z) + \sigma^2\mathbf{I}\right)^{-1} (\mathbf{y}-\hat{\mathbf{z}})\right\} \tag{45}$$

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and

$$\frac{\partial^2 g(\boldsymbol{\theta})}{\partial^2 \theta_l} = -2\Re\left\{\left(\frac{\partial^2 \hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial^2 \theta_l}\right)^{\mathrm{H}} \left(\mathrm{diag}(\boldsymbol{\mu}^z)+\sigma^2\mathbf{I}\right)^{-1}(\mathbf{y}-\hat{\mathbf{z}})\right\} + 2\left(\frac{\partial \hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial \theta_l}\right)^{\mathrm{H}} \left(\mathrm{diag}(\boldsymbol{\mu}^z)+\sigma^2\mathbf{I}\right)^{-1}\frac{\partial \hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial \theta_l}. \quad (46)$$

By the chain rule and referring to Fig. 3, we calculate $\partial\hat{\mathbf{z}}(\boldsymbol{\theta})/\partial\theta_l$ and $\partial^2\hat{\mathbf{z}}(\boldsymbol{\theta})/\partial^2\theta_l$ as

$$\frac{\partial \hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial \theta_l} = \frac{\partial \hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial \hat{\mathbf{p}}(\boldsymbol{\theta})}\frac{\partial \hat{\mathbf{p}}(\boldsymbol{\theta})}{\partial \theta_l} = \boldsymbol{\mu}^z/\boldsymbol{\mu}^p \odot \left(\hat{u}_l \frac{\partial \mathbf{Z}^{(l)}(\theta_l)}{\partial \theta_l}\hat{\mathbf{x}}\right) \quad (47)$$

and

$$\frac{\partial^2 \hat{\mathbf{z}}(\boldsymbol{\theta})}{\partial^2 \theta_l} = \boldsymbol{\mu}^z/\boldsymbol{\mu}^p \odot \left(\hat{u}_l \frac{\partial^2 \mathbf{Z}^{(l)}(\theta_l)}{\partial^2 \theta_l}\hat{\mathbf{x}}\right) \quad (48)$$

where

$$\frac{\partial \mathbf{Z}^{(l)}(\theta_l)}{\partial \theta_l} = \boldsymbol{\Gamma}^{(l)} \odot \mathbf{Z}^{(l)}(\theta_l) \quad (49)$$

$$\frac{\partial^2 \mathbf{Z}^{(l)}(\theta_l)}{\partial^2 \theta_l} = \left(\boldsymbol{\Gamma}^{(l)}\right)^{\odot 2} \odot \mathbf{Z}^{(l)}(\theta_l) \quad (50)$$

$$\boldsymbol{\Gamma}^{(l)} = \begin{bmatrix} \mathbf{0}_{l\times(N_c-l)} & \mathbf{0}_{l\times l} \\ \mathrm{diag}(\mathbf{b}) & \mathbf{0}_{(N_c-l)\times l} \end{bmatrix} \quad (51)$$

and $\mathbf{b} = [\mathbf{0}_{1\times(l-1)}, \mathrm{j}(l-1), \ldots, \mathrm{j}(N_c-1)]^{\mathrm{T}}$.

6) Calculating the Extrinsic Mean and Variance of $\mathbf{u}$ *and* $\mathbf{x}$ *in Submodule* B_3*:* The intermediate parameters $\hat{\mathbf{s}}$ and $\boldsymbol{\mu}^s$ used to compute the extrinsic means and variances of $\mathbf{u}$ and $\mathbf{x}$ are calculated as

$$\mu_m^s = \frac{1}{\mu_m^p + \sigma^2} \quad (52a)$$

$$\hat{s}_m = \frac{y_m - \hat{p}_m}{\mu_m^p + \sigma^2}. \quad (52b)$$

Then, the extrinsic variance and mean of u_l are calculated as [7]

$$\mu_l^q = \left(\sum_{m=0}^{N_c-1} \mu_m^s \left|\hat{z}_m^{(l,*)}(\theta_l)\right|^2\right)^{-1} \quad (53a)$$

$$\hat{q}_l = \hat{u}_l + \mu_l^q \sum_{m=0}^{N_c-1} \hat{s}_m \hat{z}_m^{(l,*)}(\theta_l) - \mu_l^q \hat{u}_l \sum_{m=0}^{N_c-1}\mu_m^s \sum_{j=0}^{N_c-1}\mu_j^x \left|z_m^{(l,j)}(\theta_l)\right|^2. \quad (53b)$$

Similarly, the extrinsic variance and mean of x_j are computed as

$$\mu_j^r = \left(\sum_{m=0}^{N_c-1} \mu_m^s \left|\hat{z}_m^{(*,j)}(\theta_l)\right|^2\right)^{-1} \quad (54a)$$

$$\hat{r}_j = \hat{x}_j + \mu_j^x \sum_{m=0}^{N_c-1} \hat{s}_m \hat{z}_m^{(*,j)}(\theta_l) - \mu_j^r \hat{x}_j \sum_{m=0}^{N_c-1}\mu_m^s \sum_{l=0}^{L-1}\mu_l^u \left|z_m^{(l,j)}(\theta_l)\right|^2 \quad (54b)$$

respectively. These variables are then used to calculate the L_{ext} required by LDPC decoder, i.e.,

$$L_{\mathrm{ext}}(c_{j,q}) = \ln \frac{\sum_{\alpha_n\in\mathcal{A}_q^0} \exp\left(-\frac{|\alpha_n-\hat{r}_j|^2}{\mu_j^r}\right)}{\sum_{\alpha_n\in\mathcal{A}_q^1} \exp\left(-\frac{|\alpha_n-\hat{r}_j|^2}{\mu_j^r}\right)} \quad (55)$$

where $\mathcal{A}_q^0$ and $\mathcal{A}_q^1$ are the sets of symbols, whose corresponding qth bit is 0 and 1, respectively.

B. Updating the Nuisance Parameters

Herein, we have closed the loop of the proposed algorithm when the noise variance and the nuisance parameters of the assumed prior distribution, such as σ^2, λ, ζ, and ψ are known. However, these nuisance parameters are usually unknown in practice, and one has to estimate them during iteration. We employ the EM algorithm and the nuisance parameters are updated by [27]

$$\hat{\sigma}^2 = \frac{1}{N_c}\sum_{m=0}^{N_c-1}\left(|y_m - \hat{z}_m|^2 + \mu_m^z\right) \quad (56a)$$

$$\hat{\lambda} = \frac{1}{L}\sum_{l=0}^{L-1}\pi_l \quad (56b)$$

$$\hat{\zeta} = \frac{\sum_{l=0}^{L-1}\pi_l\gamma_l}{\sum_{l=0}^{L-1}\pi_l} \quad (56c)$$

$$\hat{\psi} = \frac{1}{\lambda L}\sum_{l=0}^{L-1}\pi_l\left(|\zeta-\gamma_l|^2+\kappa_l\right). \quad (56d)$$

Herein, the whole algorithm is closed and is summarized in Algorithm 2.

C. Computational Complexity

PBiGAMP-NDS consists of the initialization and the iterative steps. For the initialization, its computation complexity is dominated by the Doppler frequencies estimation step (36) and the least squares step in Algorithm 1, whose computation complexities are $\mathcal{O}(N_p^3)$ and $\mathcal{O}(L^3)$. The number of iterations in the initialization is L, and the computation complexity of the initialization is $\mathcal{O}(N_p^3 L + L^4)$. As for the iterations of the PBiGAMP, (38b), (53b), and (54b) are the most intensive steps with computation complexity $\mathcal{O}((N_c+L)N_c)$. Let $\mathrm{Iter}_{\max}$ denote the number of iterations. In general, $N_c > N_p > L$, the computation complexity of PBiGAMP-NDS is $\mathcal{O}(N_p^3 L + N_c^2 \mathrm{Iter}_{\max})$. As for the PBiGAMP and PBiGAMP-CFO, both can be implemented efficiently in the frequency domain via the fast Fourier transform (FFT) operations, and the computation complexities of both are $\mathcal{O}(N_c \mathrm{Iter}_{\max} \log N_c)$. It can be seen that the computation complexity of PBiGAMP-NDS is higher than those of the PBiGAMP and PBiGAMP-CFO.

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Algorithm 2: PBiGAMP-NDS-Based Iterative Receiver.

Input: $\mathbf{y}$, $\max_\text{it}_{\text{turbo}}$, $\max_\text{it}_{\text{amp}}$, σ^2, λ, ζ, ψ, L_{apr}, $\hat{\mathbf{r}}$, $\boldsymbol{\mu}^r$, $\hat{\mathbf{q}}$, $\boldsymbol{\mu}^q$, $\hat{\boldsymbol{\theta}}$, L
Output: L_{pos}
Initialize: $\text{it}_{\text{turbo}} = 0$, $\text{it}_{\text{amp}} = 0$
while $\text{it}_{\text{turbo}} < \max_\text{it}_{\text{turbo}}$ **do**
  $\text{it}_{\text{turbo}} \leftarrow \text{it}_{\text{turbo}} + 1$
  while $\text{it}_{\text{amp}} < \max_\text{it}_{\text{amp}}$ **do**
    $\text{it}_{\text{amp}} \leftarrow \text{it}_{\text{amp}} + 1$
    Update posterior mean $\hat{\mathbf{u}}$ and variance $\boldsymbol{\mu}^u$ of $\mathbf{u}$ using (69)
    Update posterior mean $\hat{\mathbf{x}}$ and variance $\boldsymbol{\mu}^x$ of $\mathbf{x}$ using (74)
    Update extrinsic observation $\hat{\mathbf{p}}$ and variance $\boldsymbol{\mu}^p$ of $\mathbf{z}$ using (38)
    Update posterior mean $\hat{\mathbf{z}}$ and variance $\boldsymbol{\mu}^z$ of $\mathbf{z}$ using (77)
    Update Doppler frequencies estimates $\hat{\boldsymbol{\theta}}$ using (42)
      Update $\hat{\mathbf{s}}$ and $\boldsymbol{\mu}^s$ using (52)
    Update extrinsic observation $\hat{\mathbf{q}}$ and variance $\boldsymbol{\mu}^q$ of $\mathbf{x}$ using (53)
    Update extrinsic observation $\hat{\mathbf{r}}$ and variance $\boldsymbol{\mu}^r$ of $\mathbf{u}$ using (54)
  end while
  Update extrinsic log-likelihood ratios L_{ext} using (55)
  Using L_{ext} to calculate posterior log-likelihood ratios L_{pos} via a standard decoder
  $L_{\text{apr}} \leftarrow L_{\text{pos}} - L_{\text{ext}}$
  Update nuisance parameters σ^2, λ, ζ, ψ using (56)
end while

TABLE I
SIMULATED SC PARAMETERS SETTING

Symbol rate R_{sym}	4 kbaud
Carrier frequency f_c	15.6 kHz
Roll-off factor α_r of $g(t)$	0.22
Number of pilots N_p	255
Number of symbols N_{cd}	648
Number of zero guards N_g	40
Code rate R	1/2

IV. NUMERICAL SIMULATIONS

In this section, numerical simulations are conducted to illustrate the performance of the proposed PBiGAMP-NDS algorithm in single carrier UWA communication systems under nonuniform Doppler shifts. The carrier frequency and symbol rate are taken as $f_c = 15.6$ kHz and $R_{\text{sym}} = 4$ kBaud, respectively. The simulation parameters of the SC system are listed in Table I. The LDPC codes from IEEE 802.11ac standard [28] is adopted. The size of parity check matrix is 648×1296. The pilot sequence is an M-sequence of length 255 modulated by BPSK, and the data symbol is drawn from QPSK constellation. Benchmark algorithms are listed as follows.

1) *PBiGAMP:* The joint CE and data decoding algorithm used for the LTI system (16) with $\theta = 0$.
2) *PBiGAMP-CFO:* The joint CE and data decoding algorithm used for the LTI system (16) with a common CFO θ, where the CFO is estimated via the Newton method.
3) *PBiGAMP-initialization only (IO):* A simplified version of PBiGAMP-NDS without performing Newton refinement on the residual Doppler shifts during the iteration. It is numerically found that without the Newton refinement in the initialization, PBiGAMP-NDS performs poor. During the iteration, no Newton refinement is implemented and the algorithm is named as PBiGAMP-IO.
4) *LSMP:* The OMP-based CE methods exploiting the sparsity in the delay-Doppler domain [13].
5) *SBL:* The SBL-based CE methods exploiting the sparsity in the delay-Doppler domain [17].
6) *LSMP-based iterative receiver (LSMP-IR):* Similarly to LSMP, but after LDPC decoding, the hard decision data symbols from LDPC decoder are utilized as pilots to re-estimate the channel [29], and then the data symbols are re-estimated and decoded. This algorithm is termed as LSMP-IR.
7) *SBL-based iterative receiver (SBL-IR):* Similarly to SBL, but after LDPC decoding, the hard decision data symbols from LDPC decoder are utilized as pilots to re-estimate the channel, and then the data symbols are re-estimated and decoded. This algorithm is termed as SBL-IR.

Note that for LSMP, SBL, LSMP-IR, and SBL-IR, the MMSE-DFE turbo equalizer is used to estimate the symbols.

A. Coded BER Versus SNRs Under Random Channel Model With Different Leakage

The coded BER performance of PBiGAMP-NDS versus SNRs under random channel model with different leakage degrees is first evaluated. The variable ρ^{lk} is used to define the ratio of the energy of the equivalent noise to that of the true measurement (15), formulated as

$$\rho^{\text{lk}} = \frac{\|\mathbf{w}^s\|_2^2}{\|\mathbf{y}\|_2^2} \tag{57}$$

where the equivalent noise is $w_n^s = \sum_l w_{n,l}^{\nu} + w_n^{\text{lk}}$. Random generation is employed for channel amplitudes, delays, and Doppler rates in the following manner: there are 10 paths, the arrival time between two adjacent paths follows an exponential distribution with mean value 1 ms and the amplitude of each path follows Rayleigh distribution with mean power decreasing by 20 dB in 30 ms delay spread [11], [19]. Note that the channel length L varies in each channel realization, and it is calculated that L ranges from 17 to 117, and $N_p = 255 > L$. Considering that the Doppler rates are proportional to the relative moving speed $a_k = v_k/c$, where v_k is the relative moving speed of the kth path, v_k is supposed to follow the Gaussian distribution with zero mean and variance σ_v^2 [11]. Leakage occurs when the Doppler frequencies for two paths k and k'

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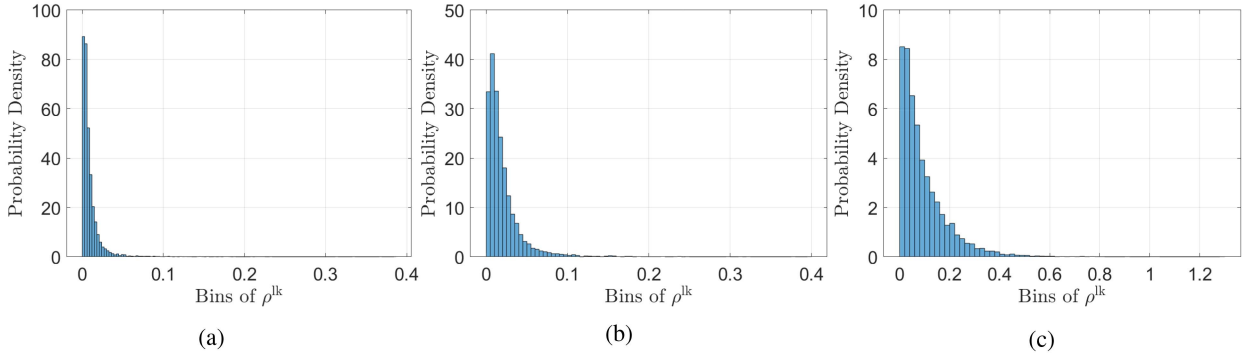


Fig. 4. Histograms of ρ^{lk} for channels with $\sigma_v = 0.1$ m/s under different N_τ: (a) $N_\tau = 2$; (b) $N_\tau = 1$; (c) $N_\tau = 0$.

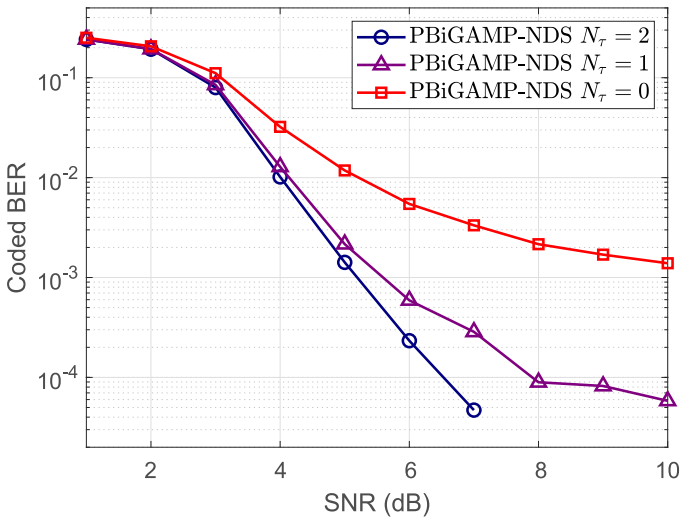


Fig. 5. Coded BER of PBiGAMP-NDS versus SNR under different leakage degrees.

differ significantly, such that $|\nu_k - \nu_{k'}| > \nu_{\mathrm{th}}$ and their delays are very close. The path delays and the Doppler frequencies are generated in two stages. First, we generate the delays of the paths. Second, for any two channel paths whose delays are close such that $|\tau_k - \tau_k'| < N_\tau T_s$, their residual Doppler frequencies are generated such that $|\nu_k - \nu_{k'}| < \nu_{\mathrm{th}}$. The above implies that once the path delays are fixed, increasing the prescribed parameter N_τ makes more paths have the similar residual Doppler frequencies, leading to a smaller leakage. We set $\nu_{\mathrm{th}} = 0.098$ rad/s, corresponding to the velocity difference 0.0015 m/s. We investigate three cases corresponding to $N_\tau = 2$, $N_\tau = 1$, and $N_\tau = 0$, with $\sigma_v = 0.1$ m/s. Specifically, $N_\tau = 0$ means that we do not impose any constraints on delays and the residual Doppler frequencies. The histograms of ρ^{lk} (57) and the coded BER of PBiGAMP-NDS under different leakage degrees are shown in Figs. 4 and 5, respectively. As N_τ decreases from 2 to 1, the leakage tends to increase and the coded BER performance degradation is small. As N_τ decreases to 0, the leakage continues to increase and the performance degradation is large.

Next, we evaluate the coded BER performances of the various algorithms under four channel settings.

1) LTI channel (Doppler shifts of all channel paths are zero).
2) LTI channel with a common CFO (Doppler shifts of all channel paths are identical and nonzero), where v is drawn i.i.d. from $\mathcal{N}(0, \sigma_v^2)$ with $\sigma_v = 0.05$ m/s.
3) Channel with the nonuniform Doppler shifts, where the velocity of each path is drawn i.i.d. from $\mathcal{N}(0, \sigma_v^2)$, where $\sigma_v = 0.05$ m/s.
4) Channel with the nonuniform Doppler shifts, where the velocity of each path is drawn i.i.d. from $\mathcal{N}(0, \sigma_v^2)$ with $\sigma_v = 0.1$ m/s.

The amplitudes and delays are generated similar to the above simulation [11], [19]. We set $N_\tau = 2$ and $\nu_{\mathrm{th}} = 0.098$ rad/s. Results are shown in Fig. 6. Fig. 6(a) shows that for LTI channel, the PBiGAMP and PBiGAMP-CFO perform best, followed by the PBiGAMP-NDS, SBL-IR, SBL, LSMP-IR, and LSMP. Compared to PBiGAMP, the performance degradation of PBiGAMP-NDS is marginal. Fig. 6(b) shows the result of the LTI channel with a common CFO where PBiGAMP-CFO performs best, followed by PBiGAMP-NDS, PBiGAMP, SBL-IR, LSMP-IR, SBL, and LSMP. Compared with the PBiGAMP-CFO, the performance degradation of PBiGAMP-NDS is still marginal. For the channel with a nonuniform Doppler shift with $\sigma_v = 0.05$ m/s and $N_\tau = 2$, the results are shown in Fig. 6(c), where PBiGAMP-NDS performs best. As SNR increases to 7 dB, PBiGAMP-IO performs better than the other algorithms except PBiGAMP-NDS. For the coded BER 10^{-3}, compared to the original PBiGAMP, the SNR gain of PBiGAMP-NDS is 2.5 dB. Fig. 6(d) shows the result when σ_v increases to 0.1 m/s, all the other algorithms except PBiGAMP-NDS and PBiGAMP-IO perform poor. For coded BER being 10^{-3}, the SNR gain of PBiGAMP-NDS is at least 2.5 dB, compared with PBiGAMP-IO, which demonstrates the benefits of performing Newton refinement in refining the Doppler frequency estimation in PBiGAMP. To sum up, the above results show that PBiGAMP-NDS is effective in handling the above four channel settings.

B. Coded BERs Versus SNRs Under the BELLHOP Channel Model

In this part, the BELLHOP beam-tracing model [30] based on the acoustic ray propagation theory, is employed to generate the channel. The simulation setups of the environments are shown in Fig. 7. In detail, the ocean depth is 25 m, the P-wave

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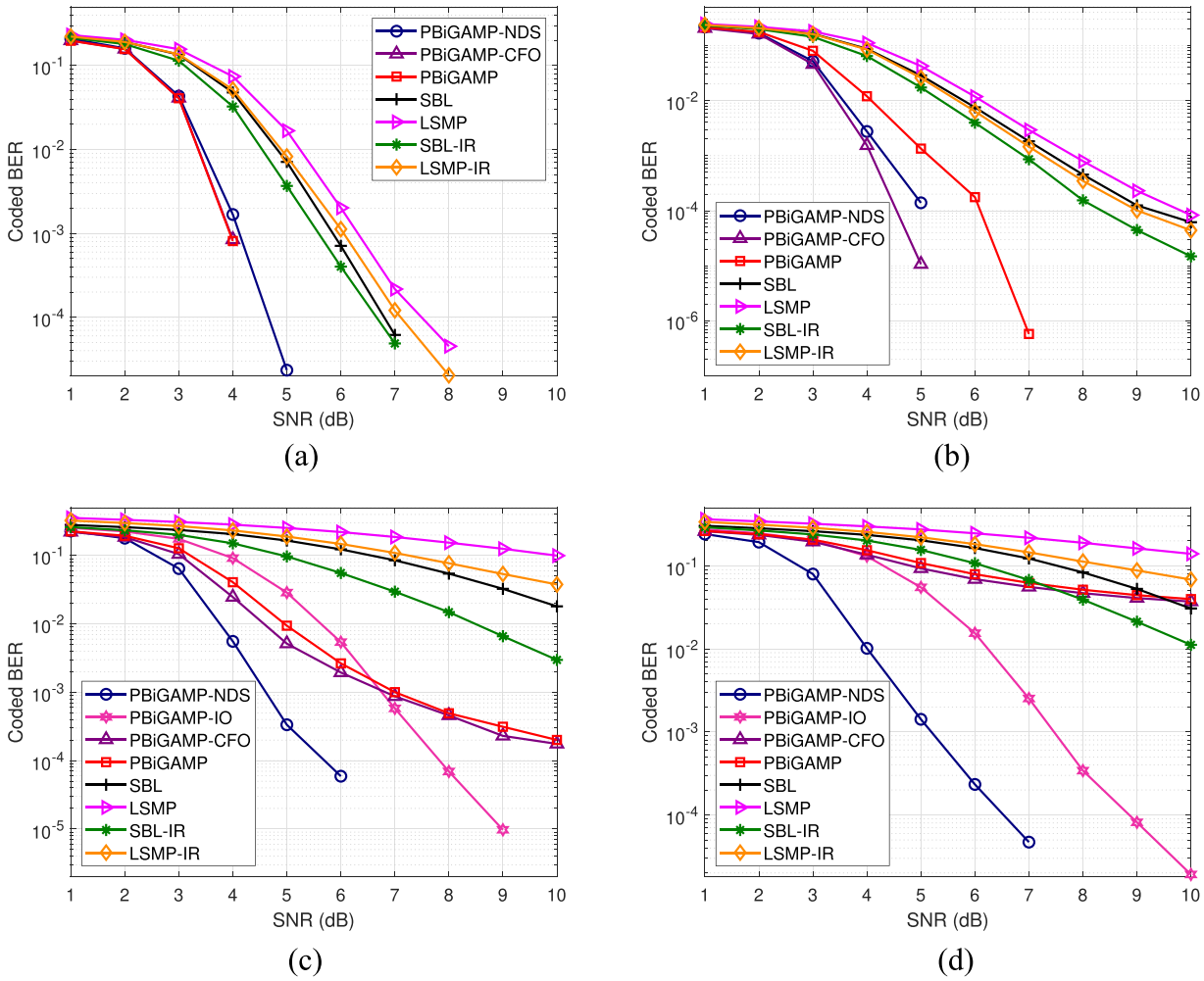


Fig. 6. Coded BER (after 1 turbo iteration) versus SNR. (a) LTI channel. (b) Channel paths have a common randomly generated Doppler shift drawn from $\mathcal{N}(0, \sigma_v^2)$ with $\sigma_v = 0.05$ m/s, which can also be regarded as a LTI channel with a CFO. (c) Each channel path has an independent randomly generated Doppler shift drawn from $\mathcal{N}(0, \sigma_v^2)$ with $\sigma_v = 0.05$ m/s. (d) Each channel path has an independent randomly generated Doppler shift with a larger $\sigma_v = 0.1$ m/s.

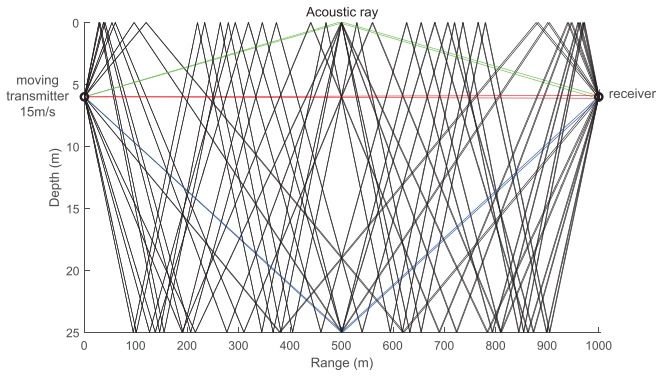


Fig. 7. Acoustic ray paths of an underwater multipath channel. The red line, the green line, the blue line, and the black line denote the direct path, one bounce surface reflected path, one bounce bottom reflected path and multiple bounce surface, and bottom reflected path, respectively.

attenuation coefficient of the bottom is 0.9 dB/λ, the speed of the sound is 1500 m/s. Both the transmitter and the receiver are fixed at 6 m depth, the transmitter is moving toward the source with a horizontal velocity $v = 15$ m/s. The range between the transmitter and the receiver is $R_0 = 1$ km. The number of calculated beams is 2000, while 18 of them that directly connect the transmitter and receiver are taken into account, as shown in Fig. 7.

Because the transmitter is moving toward the receiver, the Doppler rate of each path is positive and is calculated by [14], [31]

$$a_k = \frac{v \cos \omega_k}{c}$$

where ω_k is the ray takeoff angle at the transmitter. To visualize the equivalent narrowband channel, we assume the received signal is resampled by $\hat{a} = a_0$,[3] and show the delay-Doppler channel amplitude, i.e., the modulus of $\eta_k e^{\mathrm{j}\nu_k(lT_s+t_k)} p(t_k - lT_s)$ in Fig. 8.

The normalized signal residual error (NSRE) $\mathrm{NSRE}(\hat{\mathbf{z}})$, together with the coded BER are used as the performance criteria,

[3] Numerically we have found that the estimated Doppler rate $\hat{a}$ is very close to the Doppler rate a_0 of the first path, as the first path is often the strongest path.

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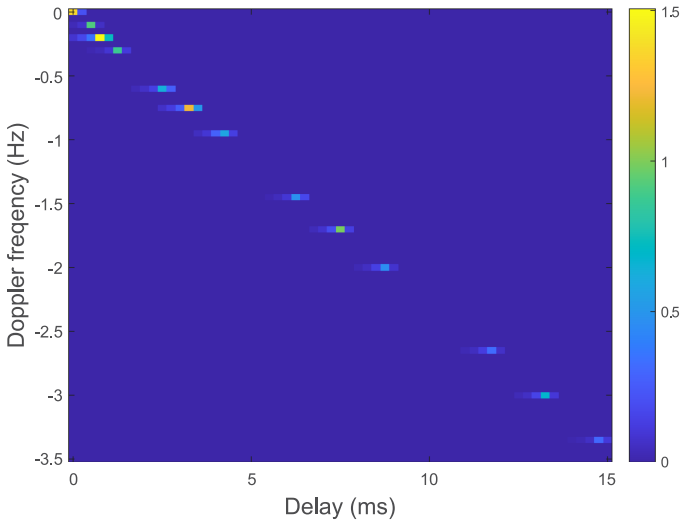


Fig. 8. Delay-Doppler channel amplitude, (the modulus of $\eta_k e^{j\nu_k(lT_s+t_k)}p(t_k-lT_s)$) obtained by BELLHOP for the environment configuration shown in Fig. 7. As the Doppler frequencies are continuous variables, they are discreted into grids with resolution being 0.05 Hz.

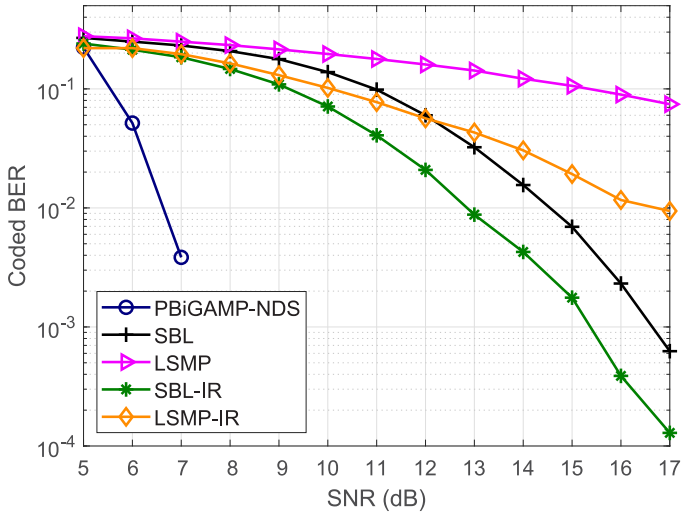


Fig. 9. Coded BERs versus the SNR for PBiGAMP-NDS, LSMP, SBL, LSMP-IR, and SBL-IR.

where $\hat{\mathbf{z}}$ denotes the signal $\mathbf{z}$ reconstructed by the estimated channel of the algorithms, and $\mathrm{NSRE}(\hat{\mathbf{z}})$ is defined as

$$\mathrm{NSRE}(\hat{\mathbf{z}}) = \frac{\|\mathbf{z}-\hat{\mathbf{z}}\|_2^2}{\|\mathbf{z}\|_2^2}.$$

For the proposed algorithm, we have $\hat{\mathbf{z}} = \sum_{l=0}^{L-1} \hat{u}_l \mathbf{Z}^{(l)}(\hat{\theta}_l)\mathbf{x}$, where $\hat{u}_l$ and $\hat{\theta}_l$ denote the estimates of the lth channel tap and the equivalent Doppler frequency, respectively, $\mathbf{x}$ is the transmitted symbol. For the benchmark algorithms, namely LSMP, SBL, LSMP-IR, and SBL-IR, we have $\hat{\mathbf{z}} = \sum_{i=0}^{I-1} \mathbf{\Phi}_i \mathbf{U}\mathbf{x}$, where I is the number of the Doppler frequency grids, $\mathbf{\Phi}_i \triangleq \mathrm{diag}([e^{j\theta_i 0}, \ldots, e^{j\theta_i(N_c-1)}])$ with θ_i being the Doppler frequency of the ith grid. $\mathbf{U}$ is a $N_c \times L$ Toeplitz matrix whose first row and first column are $[\hat{u}_{i,0}, \mathbf{0}_{L-1\times 1}]$ and $[\hat{u}_{i,0}, \ldots, \hat{u}_{i,L-1}, \mathbf{0}_{N_c-L\times 1}]^{\mathrm{T}}$, respectively, and $\hat{u}_{i,l}$ is the estimated delay-Doppler channel as defined in [12], [13], [15], [17], and [31]. The lengths of the feedforward and feedbackward filter of MMSE-DFE turbo equalizer are set as 62 and 60, respectively. As LSMP, SBL, LSMP-IR and SBL-IR are on-grid methods, the Doppler frequency grid spacing is 0.5 Hz, and the number of grids is 9 to cover the $[-4, 0.5]$ Hz Doppler span. For PBiGAMP and its variants, the maximum delay spread normalized by the symbol duration L is set as 61, the maximum number of iterations $\mathrm{max_it}_{\mathrm{turbo}}$ is set as 150. The pilots are used to estimate the common Doppler rate a and are then used to synchronize the data part. For the received baseband signal $\bar{y}(t)$ (62), given an assumed resampling factor b, according to model (63), we construct the matched filter

$$s_{\mathrm{MF}}(t) = \left(\sum_{n=0}^{N_p-1} x_n g(t-nT_s)\right)\Bigg|_{t=-t}^{*} = \sum_{n=0}^{N_p-1} x_n^* g(-t-nT_s) \tag{58}$$

where $s^*(t)$ denotes the conjugation of $s(t)$. The assumed signal $\bar{y}(t/(1+b))e^{-j2\pi f_c b(1+b)t}$ passes through the matched filter $s_{\mathrm{MF}}(t)$, and the peak of the output occurs at time $\hat{t}$ and the resampling factor $\hat{a}$, i.e.,

$$(\hat{a}, \hat{t}) = \underset{b,t}{\mathrm{argmax}} \left| \left(\left(\bar{y}\left(\frac{t}{1+b}\right) e^{-j2\pi f_c \frac{b}{1+b} t} \right) * s_{\mathrm{MF}}(t) \right) \right|. \tag{59}$$

In this way, the resampling factor $\hat{a}$ is estimated.

The coded BER performances are first investigated and results are shown in Fig. 9. PBiGAMP and PBiGAMP-CFO fail to decode the symbols. Besides, PBiGAMP-IO fails in some MC trials due to the large Doppler span. Therefore, their results are not plotted. The PBiGAMP-NDS achieves the lowest coded BER, followed by SBL-IR, SBL, LSMP-IR, and LSMP. Compared with other benchmark algorithms, PBiGAMP-NDS performs significantly better, which demonstrates the effectiveness of the established channel model, the advantages of the gridless Doppler frequency estimation and the joint CE and data decoding.

Fig. 10(a)–10(c) displays, respectively, the estimated delay-Doppler channel of LSMP-IR, SBL-IR, and PBiGAMP-NDS at SNR = 17 dB. It can be observed that all three methods estimate the channel's delay and Doppler parameters well. However, since LSMP-IR and SBL-IR are on-grid approaches, their Doppler estimates are inherently constrained to the predefined grid points. By contrast, PBiGAMP-NDS produces Doppler frequency estimates that match the ground truth closely at almost all delay bins.

Fig. 11 shows the NSREs per sample over the entire package duration, i.e., $\mathrm{NSRE}(\hat{z}_m) \triangleq |z_m - \hat{z}_m|^2/|z_m|^2$. It can be seen that in the pilot part (the first 255 symbols), the NSREs of SBL-IR, PBiGAMP-NDS are similar. However, in the data part, PBiGAMP-NDS performs better than SBL-IR and LSMP-IR.

Lastly, Fig. 12 plots the $\mathrm{NSRE}(\hat{\mathbf{z}})$ versus the SNR for the PBiGAMP-NDS, LSMP, SBL, LSMP-IR, and SBL-IR. The

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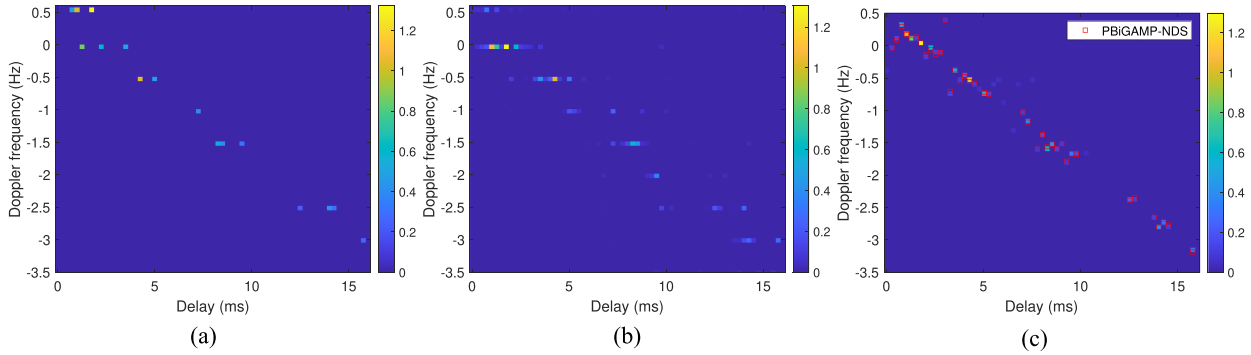


Fig. 10. Estimated delay-Doppler channel of the three algorithms (a) LSMP-IR, (b) SBL-IR, and (c) PBiGAMP-NDS (PBiGAMP-NDS yields gridless Doppler frequencies estimates, which are rounded to their nearest Doppler grids). The red square markers denote the gridless estimates of the PBiGAMP-NDS without showing their amplitude. All the results are obtained at SNR $=$ 17 dB.

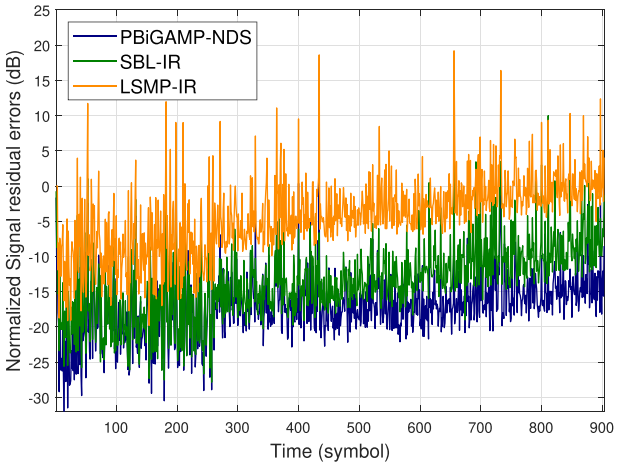


Fig. 11. $\mathrm{NSRE}(\hat{z}_m)$ versus the time (normalized by symbol duration) at SNR $=$ 14 dB, obtained by LSMP-IR, SBL-IR, and PBiGAMP-NDS.

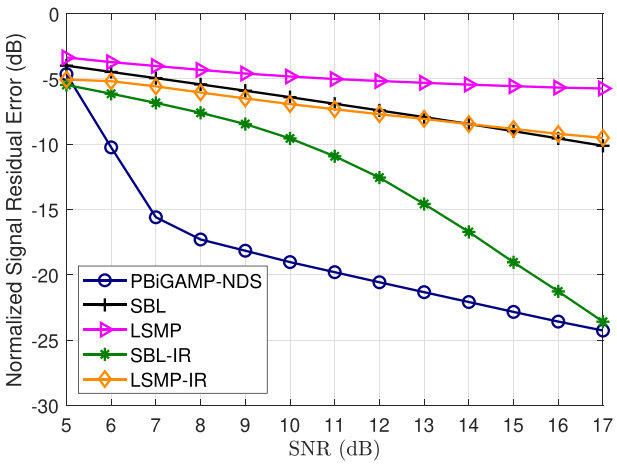


Fig. 12. $\mathrm{NSRE}(\hat{\mathbf{z}})$ versus SNR of the LSMP, SBL, LSMP-IR, SBL-IR, and the proposed PBiGAMP-NDS.

PBiGAMP-NDS achieves the lowest NSRE, followed by SBL-IR, SBL, LSMP-IR and LSMP. The NSERs of LSMP-IR and SBL are similar. This indicates that the approximated established model approximates the true input–output model very accurately

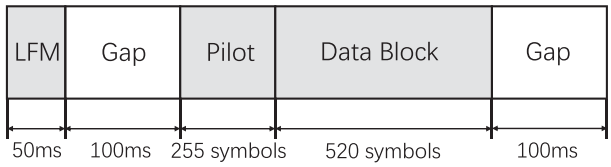


Fig. 13. Transmitted package structure used in the sea trial.

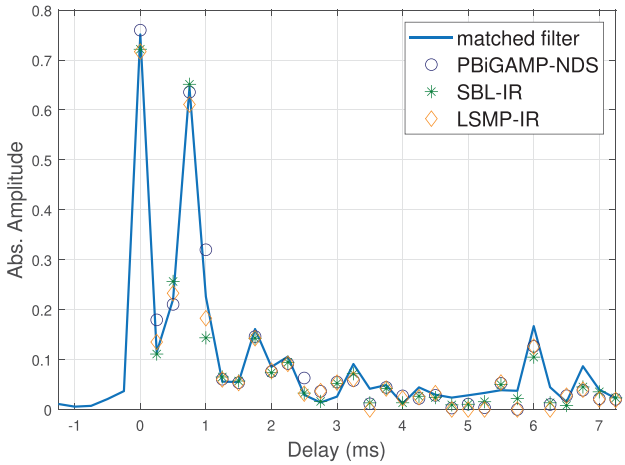


Fig. 14. Channel amplitudes obtained by matched filter, PBiGAMP-NDS, SBL-IR, and LSMP-IR. For SBL-IR and LSMP-IR, the amplitude corresponding to the lth delay grid is to sum up the channel amplitudes across all Doppler grids associated with that delay grid, i.e., $\sum_{i=0}^{I-1} \hat{u}_{i,l}$. The 4th data package is used to obtain the results.

in the simulated channel setup, performing the gridless Doppler frequency benefits the CE significantly, and performing the joint CE and data detection brings some performance gains especially at lower SNRs.

V. Sea Trial Data Decoding Results

To verify the performance of the proposed PBiGAMP-NDS approach in real UWA channels, data from shallow-water acoustic communication experiment are analyzed, and the results are delineated. The experiment was conducted off the coast of Zhoushan, China on July 26, 2023. The sea depth is 25 m. The receiver was released from a vessel drifting at 2 m/s due

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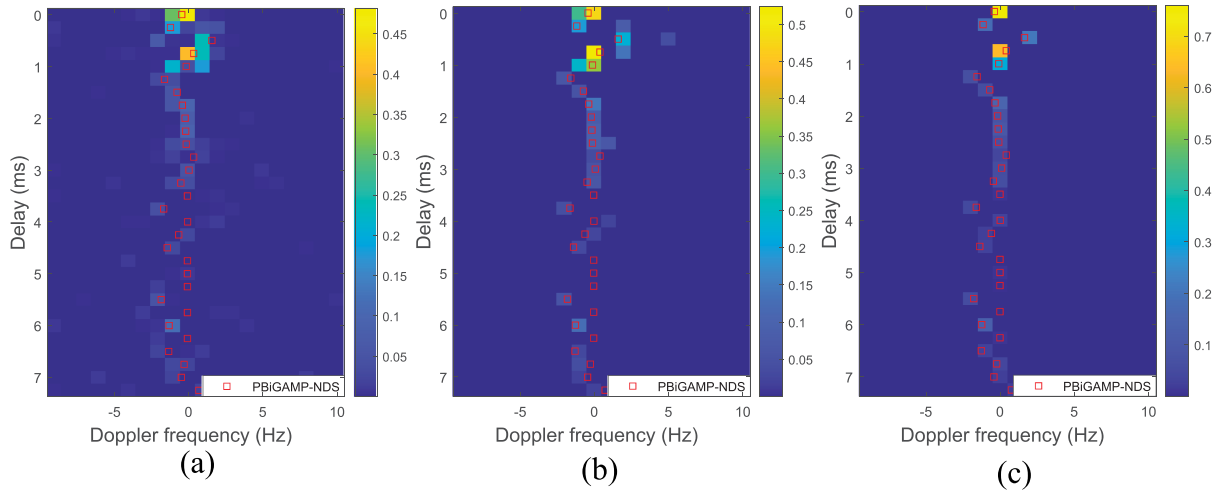


Fig. 15. Estimated delay-Doppler channel of the three algorithms (a) SBL-IR, (b) LSMP-IR, and (c) PBiGAMP-NDS. The red square markers denote the gridless estimates of the PBiGAMP-NDS without showing their amplitudes. All the results are obtained via processing the 4th data package.

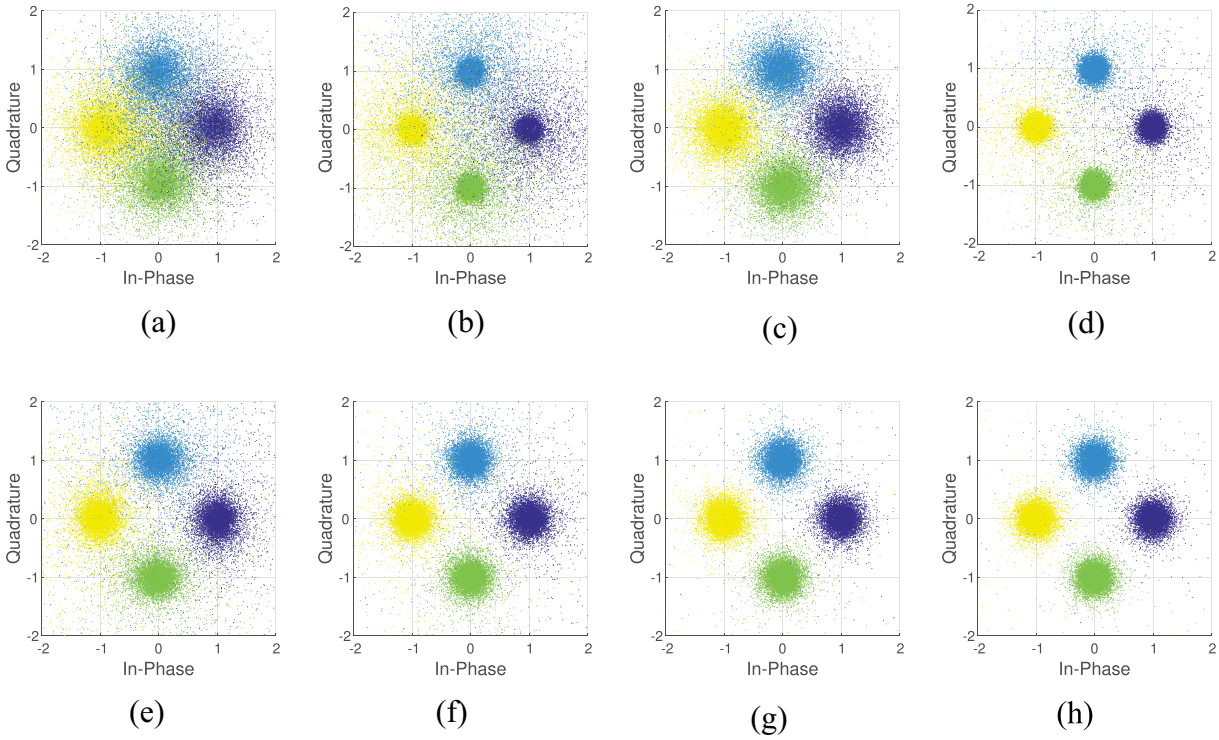


Fig. 16. Symbol estimates of all the 64 packages after one turbo iteration (a) LSMP, (b) LSMP-IR, (c) SBL, (d) SBL-IR, (e) PBiGAMP, (f) PBiGAMP-CFO, (g) PBiGAMP-IO, and (h) PBiGAMP-NDS.

to current, while the transmitter was released from a dock at Zhairuoshan island. The depths of both the transmitter and the receiver were 5 m. The distance between them ranges from 1000 to 1500 m. In total, 64 packages corresponding to 33 280 symbols were transmitted during the experiment. For each package, the pilots are the same as in the simulation part, the data bits are coded using LDPC code adhering to 5G NR standard [32], with the size of parity check matrix being 520×1040. The coded bits are modulated by QPSK constellation. Specifically, 32 packages were transmitted over a distance of 1000 m, and the remaining 32 packages were sent over a distance of 1500 m. A raised cosine pulse-shaping filter with a roll-off factor of 0.2 is adopted, and the carrier frequency is 15.6 kHz. The entire package structure is illustrated in Fig. 13. To make performance comparison, seven benchmark algorithms, namely, LSMP, SBL, LSMP-IR, SBL-IR, PBiGAMP [6], PBiGAMP-CFO, and PBiGAMP-IO are implemented. For LSMP, SBL, LSMP-IR, and SBL-IR, the Doppler frequencies are discreted into $I = 20$ grids with grid

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TABLE II
CODED BER AND NMSES OF SYMBOL ESTIMATES IN THE SEA TRIAL

Alg.	LSMP	SBL	LSMP-IR	SBL-IR	PBiGAMP	PBiGAMP-CFO	PBiGAMP-IO	PBiGAMP-NDS
BER	0.092278	0.018780	0.053891	0.013912	0.036463	0.013221	0.004207	0.003365
NMSE	−3.99 dB	−6.87 dB	−5.88 dB	−12.34 dB	−5.34 dB	−8.54 dB	−11.00 dB	−12.05 dB

spacing being 1 Hz, and the channel estimates are input into the MMSE-DFE to perform symbol estimation. The maximum delay spread L is set as 30 for all the algorithms.

Fig. 14 shows the CIR estimates of matched filter, PBiGAMP-NDS, SBL-IR, and LSMP-IR. The channel estimates of the matched filter is obtained by convolving the received baseband signal with the time-reversed conjugation of the pilot sequence. All three methods accurately capture the taps with relatively large amplitude. Fig. 15 shows the delay-Doppler channel estimates obtained by SBL-IR, LSMP-IR, and PBiGAMP-NDS. It can be seen that the accuracy of the delay-Doppler frequency estimates with large amplitude of the three algorithms are comparable. Both SBL-IR and LSMP-IR output multiple Doppler estimates for some delay bins, while the PBiGAMP-NDS only outputs a single Doppler estimate for each delay bin. This demonstrates that PBiGAMP-NDS yields more sparse estimates.

In the following, we present the demodulated symbols after one turbo equalization, as shown in Fig. 16. The coded BERs and the NMSEs of symbol estimates are given in Table II. The PBiGAMP-NDS performs significantly better in terms of BERs than PBiGAMP-CFO, SBL-IR, SBL, PBiGAMP, LSMP-IR, and LSMP, and exhibits a slight superiority over PBiGAMP-IO. Surprisingly, the PBiGAMP-NDS algorithm is slightly inferior to SBL-IR in terms of symbol NMSE.

Additional analysis of the decoding performance, as a function of the number of turbo iterations, is depicted in Fig. 17. We observe that the uncoded BER performance of PBiGAMP-NDS is significantly lower than that of the other benchmark algorithms. Approximately half of the data packages are successfully demodulated by PBiGAMP-NDS without performing LDPC decoding, while the success rates of perfect decoding without LDPC decoding of the other algorithms are very low.

Moreover, the average $\mathrm{NSRE}(\hat{z}_m)$ of PBiGAMP-NDS, SBL-IR, and LSMP-IR are shown in Fig. 18. In the pilot part, SBL-IR achieves the lowest NSRE, followed by LSMP-IR and PBiGAMP-NDS. In the data part, the NSREs of PBiGAMP-NDS and SBL-IR are quite similar, and both are lower than that of LSMP-IR.

Finally, the processing time of the algorithms is also evaluated. The average processing time per package of the SBL and LSMP is 2.07 and 0.09 s, respectively, followed by the MMSE-DFE taking 0.57 s for one turbo iteration per package, while PBiGAMP-NDS (including initialization) with one turbo iteration per package takes 2.01 s. For SBL-IR and LSMP-IR, utilizing the hard-decision symbols to re-estimate the channel take additional 2.9 and 0.13 s, respectively. Accelerated by FFT, PBiGAMP and PBiGAMP-CFO only take 0.17 and 0.19 s per package, respectively. Compared to other methods, the coded BER performance of PBiGAMP-NDS is better, through the running time of PBiGAMP-NDS is higher than that of others, except SBL and SBL-IR.

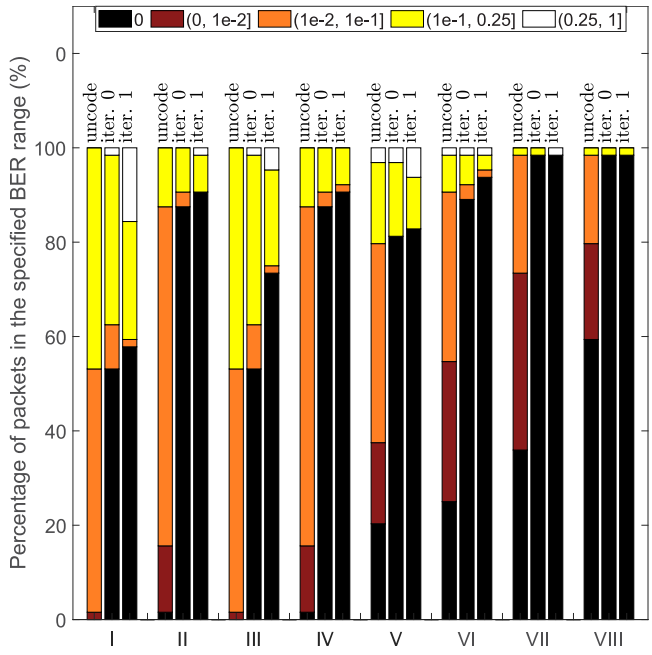


Fig. 17. Results from the sea trial as a function of the number of turbo iterations. I: LSMP, II: SBL, III: LSMP-IR, IV: SBL-IR, V: PBiGAMP, VI: PBiGAMP-CFO, VII: PBiGAMP-IO, VIII: PBiGAMP-NDS.

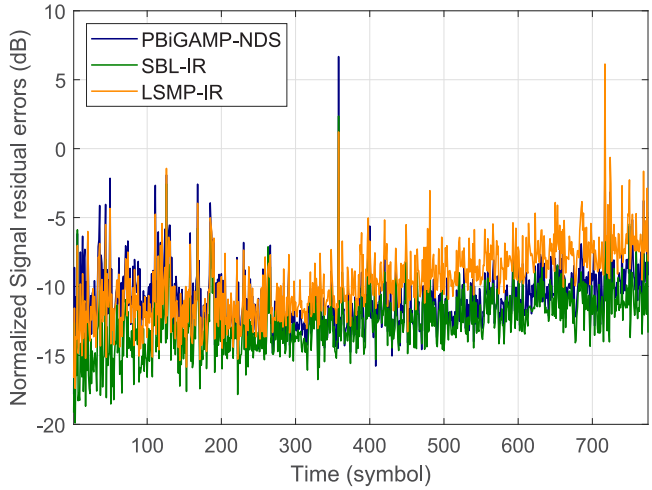


Fig. 18. Average $\mathrm{NSRE}(\hat{z}_m)$ versus the time (normalized by symbol duration) averaged over 64 packages, obtained by LSMP-IR, SBL-IR, and PBiGAMP-NDS.

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VI. Conclusion

This article establishes an input–output channel model for the UWA channel under nonuniform Doppler shifts and analyzes it in depth. A PBiGAMP-NDS single carrier iterative receiver for joint channel amplitude estimation, gridless Doppler frequency estimation, and data detection is developed. The proposed iterative receiver inherits the high CE accuracy advantages of the PBiGAMP. In addition, the iterative receiver can also be easily extended to handle the LTI channel model with a common CFO. Numerical simulations and sea trials data are used to demonstrate the effectiveness of the proposed iterative receiver, compared with LSMP, LSMP-IR, SBL, SBL-IR, PBiGAMP, and its variants.

Appendix A
Baseband Received Signal Model

The received signal undergoes the UWA channel is

$$\tilde{y}(t) = \sum_{k=0}^{p_0-1} \tilde{\eta}_k \Re\left\{ \sum_{n=0}^{N_c-1} x_n g(t-\tau_k(t)-nT_s) \mathrm{e}^{\mathrm{j}2\pi f_c(t-\tau_k(t))} \right\} + \tilde{w}(t). \tag{60}$$

Substituting and (4) and (5) in (60), $\tilde{y}(t)$ can be simplified as

$$\tilde{y}(t) = \sum_{k=0}^{p_0-1} \tilde{\eta}_k \Re\left\{ \sum_{n=0}^{N_c-1} x_n g((1+a+\delta a_k)t-\tau_k - nT_s) \mathrm{e}^{\mathrm{j}2\pi f_c((1+a+\delta a_k)t-\tau_k)} \right\} + \tilde{w}(t). \tag{61}$$

After downcoverting and low pass filtering, the baseband signal can be simplified to be

$$\begin{aligned} \bar{y}(t) &= \mathrm{LPF}\left\{ \tilde{y}(t) \mathrm{e}^{-\mathrm{j}2\pi f_c t} \right\} \\ &= \sum_{k=0}^{p_0-1} \tilde{\eta}_k \sum_{n=0}^{N_c-1} x_n g((1+a+\delta a_k)t-\tau_k \\ &\quad - nT_s) \mathrm{e}^{\mathrm{j}2\pi f_c((a+\delta a_k)t-\tau_k)} + \bar{w}(t). \end{aligned} \tag{62}$$

Note that in general, the bandwidth of $g(t)$ is smaller than the carrier frequency f_c. Thus, it is believed that resampling $\bar{y}(t)$ will eliminate the waveform compression or dilation caused by the Doppler effect. Let $\hat{a}$ denote the estimated common Doppler rate. To mitigate the wideband Doppler effect, the baseband signal is resampled by an estimated Doppler rate $\hat{a}$ and multiplied by a complex exponential signal at the frequency $-f_c\hat{a}/(1+\hat{a})$, yielding

$$\begin{aligned} y(t) &= \bar{y}\left(\frac{t}{1+\hat{a}}\right) \mathrm{e}^{-\mathrm{j}2\pi f_c \frac{\hat{a}}{1+\hat{a}} t} \\ &= \sum_{k=0}^{p_0-1} \tilde{\eta}_k \sum_{n=0}^{N_c-1} x_n g\left(\frac{1+a+\delta a_k}{1+\hat{a}} t - \tau_k - nT_s\right) \\ &\quad \mathrm{e}^{\mathrm{j}2\pi f_c\left(\frac{a+\delta a_k-\hat{a}}{1+\hat{a}} t-\tau_k\right)} + \bar{w}\left(\frac{t}{1+\hat{a}}\right) \mathrm{e}^{-\mathrm{j}2\pi f_c \frac{\hat{a}}{1+\hat{a}} t}. \end{aligned} \tag{63}$$

Suppose the estimated Doppler rate $\hat{a}$ is close to a, and $|\delta a_k|$ is usually a small value in UWA channel, $(1+a+\delta a_k)/(1+\hat{a})$ is close to 1, yielding

$$g\left(\frac{1+a+\delta a_k}{1+\hat{a}} t - \tau_k - nT_s\right) \approx g(t-\tau_k-nT_s). \tag{64}$$

Adopting the above approximation, (63) can be simplified to be

$$y(t) = \sum_{k=0}^{p_0-1} \tilde{\eta}_k \sum_{n=0}^{N_c-1} x_n g\left(t-\tau_k-nT_s\right) \mathrm{e}^{\mathrm{j}2\pi f_c\left(\frac{a+\delta a_k-\hat{a}}{1+\hat{a}} t-\tau_k\right)} + \widetilde{w}(t) \tag{65}$$

where $\widetilde{w}(t)$ denotes the equivalent noise term consisting of the approximation error in (64) and the noise $\bar{w}$ in (63). By defining η_k (7a) and ν_k (7b), (65) can be simplified as (6).

Appendix B
Denoising of $\mathbf{u}$, $\mathbf{x}$, and $\mathbf{z}$

Given the extrinsic mean $\hat{q}_l$ and variance μ_l^q, the pseudomeasurement model

$$\hat{q}_l = u_l + v_l^q \tag{66}$$

can be constructed, where v_l^q is the pseudonoise following $\mathcal{CN}(v_l^q|0,\mu_l^q)$. By combing the prior information $p(u_l;\lambda,\zeta,\psi)$ and the likelihood, which can be obtained according to (66), we perform a denoising step, i.e., the posterior mean and variance of u_l are calculated. The posterior distribution of u_l is given by [27]

$$\begin{aligned} p(u_l|\mathbf{y},\hat{q}_l,\mu_l^q) &\propto \mathcal{CN}(u_l|\hat{q}_l,\mu_l^q)p(u_l) \\ &= (1-\pi_l)\delta(u_l) + \pi_l \mathcal{CN}(u_l|\gamma_l,\kappa_l) \end{aligned} \tag{67}$$

where

$$\kappa_l = \left(\frac{1}{\mu_l^q} + \frac{1}{\psi}\right)^{-1} \tag{68a}$$

$$\gamma_l = \kappa_l \left(\frac{\hat{q}_l}{\mu_l^q} + \frac{\zeta}{\psi}\right) \tag{68b}$$

$$\pi_l = \left(1 + \frac{(1-\lambda)\mathcal{CN}(0|\hat{q}_l,\mu_l^q)}{\lambda \mathcal{CN}(0|\zeta-\hat{q}_l,\psi+\mu_l^q)}\right). \tag{68c}$$

Consequently, the posterior mean and variance of u_l are

$$\hat{u}_l = \pi_l \gamma_l \tag{69a}$$

$$\mu_l^u = \pi((1-\pi)|\gamma_l|^2 + \kappa_l^2). \tag{69b}$$

Given the decoupled extrinsic mean $\hat{r}_j$ and variance μ_j^r, the pseudomeasurement model

$$\hat{r}_j = x_j + v_j^x \tag{70}$$

can be constructed similarly, with the pseudonoise v_j^x obeying Gaussian distribution $v_j^x \sim \mathcal{CN}(v_j^x|0,\mu_j^x)$. As a coded system, the prior information $p(x_j)$ for $j \in \mathcal{D}$ that required by the denoising step is provided by the LDPC decoder in the form of LLR. Therefore, we first convert the LLRs to the prior information that we need, i.e.,

$$p(x_j = \alpha_n) = \prod_{q=0}^{Q-1} p(c_{j,q} = \chi_{n,q}) \tag{71}$$

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with

$$p(c_{j,q}) = \frac{\exp\left((-1)^{c_{j,q}} L_{\text{apr}}\left(c_{j,q}\right)/2\right)}{\exp\left(L_{\text{apr}}\left(c_{j,q}\right)/2\right) + \exp\left(-L_{\text{apr}}\left(c_{j,q}\right)/2\right)} \quad (72)$$

where $L_{\text{apr}}(c_{j,q})$ denotes the prior LLR of bit $c_{j,q}$ from the decoder, and $\chi_{n,q}$ is the qth bit corresponding to the symbol α_n. Combing the likelihood (70) and the prior PMF (71) of x_j, the posterior PMF of x_j is

$$p(x_j = \alpha_n|\mathbf{y}) \propto \mathcal{CN}(x_j = \alpha_n|\hat{r}_j, \mu_j^r)p(x_j = \alpha_n). \quad (73)$$

The posterior means and variances of $\mathbf{x}$ are input into the bilinear inference module B_3, which are calculated as

$$\hat{x}_j = \sum_{n=0}^{2^Q-1} p(x_j = \alpha_n|\mathbf{y})\alpha_n \quad (74a)$$

$$\mu_j^x = \sum_{n=0}^{2^Q-1} p(x_j = \alpha_n|\mathbf{y})|\alpha_n - \hat{x}_j|^2. \quad (74b)$$

As for the pilots and zero guard, we have $\hat{x}_j = x_j$, $\mu_j^x = 0$ for $j \in \mathcal{P} \bigcup \mathcal{G}$.

In the denoising process of $\mathbf{z}$, we approximate the prior distribution of $\mathbf{z}$ as a Gaussian distribution with mean and covariance matrix being $\hat{\mathbf{p}}$ and $\mathrm{diag}(\boldsymbol{\mu}^p)$, respectively. For the measurement model

$$y_m = z_m + w_m \quad (75)$$

its likelihood $p(y_m|z_m) = \mathcal{CN}(y_m|z_m, \sigma^2)$ can be obtained. Consequently, the posterior distribution of z_m is Gaussian distributed shown as

$$z_m \sim \mathcal{CN}(z_m|\hat{z}_m, \mu_m^z) \quad (76)$$

where the posterior variance μ_m^z and mean $\hat{z}_m$ of z_m are

$$\mu_m^z = \left(\frac{1}{\mu_m^p} + \frac{1}{\sigma^2}\right)^{-1} \quad (77a)$$

$$\hat{z}_m = \mu_m^z\left(\frac{\hat{p}_m}{\mu_m^p} + \frac{y_m}{\sigma^2}\right) \quad (77b)$$

respectively.

Appendix C
Brief Implementation of PBiGAMP-CFO

In the special case where all channel taps have the identical Doppler frequency $\tilde{\theta}$, model (19) can be rewritten as

$$\tilde{\mathbf{y}}(\tilde{\theta}) \triangleq \mathbf{e}(-\tilde{\theta}) \odot \mathbf{y} = \mathbf{U}\mathbf{x} + \mathbf{e}(-\tilde{\theta}) \odot \mathbf{w} \quad (78)$$

which is equivalent to its frequency domain version

$$\begin{aligned}\bar{\mathbf{y}}(\tilde{\theta}) &= \mathbf{F}_{N_c}\tilde{\mathbf{y}}(\tilde{\theta})/\sqrt{N_c} = \mathbf{F}_{N_c}\mathbf{U}\mathbf{x}/\sqrt{N_c} \\ &\quad + \mathbf{F}_{N_c}\mathbf{e}(-\tilde{\theta}) \odot \mathbf{w}/\sqrt{N_c} \\ &= \mathbf{F}_{N_c}\mathbf{U}\mathbf{x}/\sqrt{N_c} + \bar{\mathbf{w}}\end{aligned} \quad (79)$$

where $\bar{\mathbf{w}} \sim \mathcal{CN}(\bar{\mathbf{w}}|0, \sigma^2/N_c\mathbf{I}_{N_c})$, $\mathbf{F}_{N_c}$ is an $N_c \times N_c$ discrete Fourier transform (DFT) matrix. Actually, when $\tilde{\theta}$ is fixed, estimating $\mathbf{u}$ and $\mathbf{x}$ can be solved by the FFT accelerated PBiGAMP algorithm [33]. When $\tilde{\theta}$ is initialized and fixed as $\hat{\theta}'$ during the current iteration, the FFT accelerated PBiGAMP will provide the posterior mean and variance of $\bar{\mathbf{z}} \triangleq \mathbf{F}_{N_c}\mathbf{U}\mathbf{x}/\sqrt{N_c}$, i.e., $\hat{\mathbf{z}}$ and $\boldsymbol{\nu}^z$, respectively, see [33, (B6, 7)]. Recall that $\bar{\mathbf{y}}(\tilde{\theta}) = \bar{\mathbf{z}} + \bar{\mathbf{w}}$, and $\bar{\mathbf{w}}$ is independent of $\mathbf{z}$, one has

$$\bar{\mathbf{y}}(\tilde{\theta}) \sim \mathcal{CN}(\bar{\mathbf{y}}(\tilde{\theta})|\hat{\mathbf{z}}, (\mathrm{diag}(\boldsymbol{\nu}^z) + \sigma^2/N_c\mathbf{I}_{N_c})^{-1}). \quad (80)$$

The type II approach can be adopted to estimate $\tilde{\theta}$, where the estimation problem is formulated as

$$\underset{\tilde{\theta}}{\text{minimize}} \left(\bar{\mathbf{y}}(\tilde{\theta}) - \hat{\mathbf{z}}\right)^{\mathrm{H}} (\mathrm{diag}(\boldsymbol{\nu}^z) + \sigma^2/N_c\mathbf{I}_{N_c})^{-1}\left(\bar{\mathbf{y}}(\tilde{\theta}) - \hat{\mathbf{z}}\right) \quad (81)$$

which is equivalent to

$$\begin{aligned}&\underset{\tilde{\boldsymbol{\theta}}}{\text{minimize}} -\Re\left\{\hat{\mathbf{z}}^{\mathrm{H}}(\mathrm{diag}(\boldsymbol{\nu}^z) + \sigma^2/N_c\mathbf{I}_{N_c})^{-1}\mathbf{F}_{N_c}(\mathbf{e}(-\tilde{\theta}) \odot \mathbf{y})\right\} \\ &\triangleq g(\tilde{\theta}).\end{aligned} \quad (82)$$

Since the residual CFO after resampling and CFO compensation is usually a small value, one can adopt Newton method to solve the above problem, i.e.,

$$\hat{\theta} = \hat{\theta}' - \frac{\partial g(\tilde{\theta})}{\partial \tilde{\theta}} \Big/ \frac{\partial^2 g(\tilde{\theta})}{\partial^2 \tilde{\theta}}\Big|_{\tilde{\theta}=\hat{\theta}'} \quad (83)$$

where

$$\begin{aligned}&\frac{\partial g(\tilde{\theta})}{\partial \tilde{\theta}} = \\ &-\Re\left\{\hat{\mathbf{z}}^{\mathrm{H}}(\mathrm{diag}(\boldsymbol{\nu}^z) + \sigma^2/N_c\mathbf{I}_{N_c})^{-1}\left(\mathbf{F}_{N_c}\frac{\partial \mathbf{e}(-\tilde{\theta})}{\partial \tilde{\theta}}\mathbf{y}/\sqrt{N_c}\right)\right\}\end{aligned} \quad (84)$$

and

$$\begin{aligned}&\frac{\partial^2 g(\tilde{\theta})}{\partial^2 \tilde{\theta}} = \\ &-\Re\left\{\hat{\mathbf{z}}^{\mathrm{H}}(\mathrm{diag}(\boldsymbol{\nu}^z) + \sigma^2/N_c\mathbf{I}_{N_c})^{-1}\left(\mathbf{F}_{N_c}\frac{\partial^2 \mathbf{e}(-\tilde{\theta})}{\partial^2 \tilde{\theta}}\mathbf{y}/\sqrt{N_c}\right)\right\}.\end{aligned} \quad (85)$$

After refining $\hat{\theta}'$ as $\hat{\theta}$, $\bar{\mathbf{y}}(\hat{\theta})$ is also refined and passed into the next iteration of FFT accelerated PBiGAMP.

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